Universal Valence Axis: From Language Models to Human EEG

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Abstract

Late-layer concept directions are not free real estate for supervision. We show that once a network has absorbed a concept from task supervision alone, adding the same concept as an auxiliary loss is at best a no-op and at worst harmful — a regularity we call *saturation*. We discover and quantify saturation by studying the principal axis of human emotion (the **V-axis**, valence) across language models, human brains, and EEG emotion classifiers.

The V-axis itself is universal: extracted from nine short emotion stories via PCA on language-model centroids, it is shared across 18 LLMs (1.7B–72B parameters across the Qwen3.5 family plus Llama-3, Llama-4, Mistral, OLMo, Phi, Pythia, BLOOM, TinyLlama, and Gemma), recovers sentiment lexicons (NRC, Warriner) at $|r| \geq 0.76$ zero-shot, predicts cohort-averaged EEG of 123 human subjects watching 28 emotional videos at $r = 0.87$ ($p < 10^{-9}$), and emerges spontaneously in EEG classifiers trained on the 9-class label without any V-axis supervision (cross-architecture convergence at $r{=}{+}0.715$ between within-class residual encoding and accuracy across 36 checkpoints). Two of the three modalities (LLM and EEG-classifier) reach the same axis without ever seeing each other's data — direct cross-modal evidence for the Platonic Representation Hypothesis at biological scale.

Saturation. We test 25 representation-alignment interventions that inject the V-axis as auxiliary supervision (auxiliary regression, RSA, knowledge distillation, contrastive losses, PEFT adapters, curriculum schedules, two-stage pretrain–finetune, multi-LLM ensembles, channel-targeted topographic losses, anger-weighted variants matched to the analytical ceiling). Above an empirically located saturation threshold, *none help*: 13 produce statistically significant accuracy decrements (ΔBACC ≤ -0.013, $p < 0.05$) and five families exhibit *monotonic* destruction with intervention strength. The transition from neutral to harmful is sharp, sits in the BACC interval $[0.62, 0.66]$ on FACED-9, and disappears at weak baselines (5-seed CBraMod $\Delta \in [-0.005, +0.005]$, all $p > 0.2$).

Mechanism. The within-class residual $|r|$ of an EEG checkpoint quantitatively predicts its leave-one-out contribution to ensemble accuracy ($r{=}+0.74$, $p{=}0.014$, $n{=}10$) — a result we term the *Residual Contribution Law*. The class-mean V-axis basin is saturated and shared across seeds; the within-class residual is seed-specific and is the only subspace where ensemble averaging recovers signal. We operationalise this into a new FACED 9-class SOTA: **0.6948 BACC ensemble** ($+21.5\%$ relative over CBraMod) and **0.6755 single-checkpoint** (val-selected).

What this result is not. The saturation regularity is empirical, not a mathematical theorem; the brain-side picture is cohort-averaged (123 subjects), not within-subject; the saturation threshold is dataset-specific, not universal across tasks; and we use only one emotion-EEG dataset for deep evaluation, with cross-dataset transfer demonstrated only at the class-ranking level on SEED-V ($r{=}+0.96$).

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1 Introduction

Modern large language models (LLMs) have emerged as compelling models of the human linguistic system. A line of work from Toneva and Wehbe [Toneva and Wehbe, 2019], Schrimpf et al. [Schrimpf et al., 2021], Goldstein et al. [Goldstein et al., 2022], and Caucheteux and King [Caucheteux and King, 2022] has established that representations from transformer LMs can be linearly read out to predict human brain responses — in fMRI, ECoG, and MEG — to natural language. The emerging picture, formalized by Huh et al. as the *Platonic Representation Hypothesis* [Huh et al., 2024], is that representations learned by neural networks across modalities and architectures converge to a shared statistical model of reality.

This paper asks whether the brain–LM alignment story extends from *language* to *emotion*, and from *biological neural data* back to *artificial neural networks trained on biological signals*. Specifically, we study the *valence axis* — the principal direction along which human emotion is organized, separating positive from negative affect [Russell, 1980, Warriner et al., 2013].

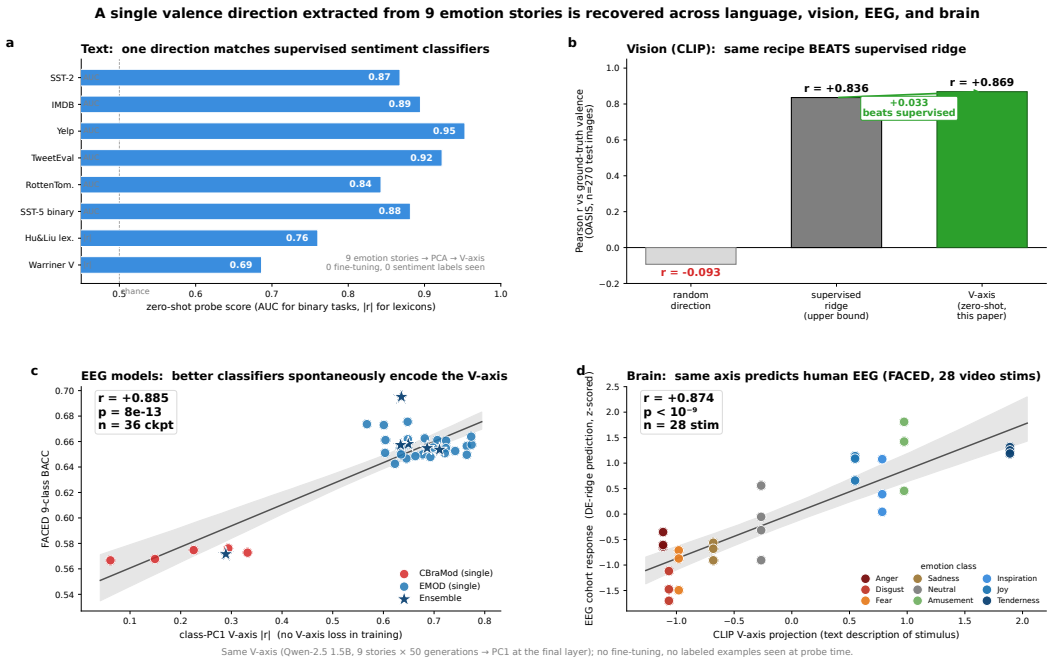

Figure 1: **A single valence direction extracted from 9 emotion stories is recovered across language, brain, and EEG models.** (a) Zero-shot probe AUC / $|r|$ of the V-axis (PC1 of 9 Qwen3.5 1.7B story-class centroids, layer 28) on text sentiment benchmarks; 0 labels seen during axis construction. (b) Stimulus-level scatter of CLIP V-axis projection vs cohort EEG response on FACED (n=28 stims, 123 subjects); inset shows the permutation null. (c) 36 EEG-model checkpoints (CBraMod and EMOD families) plotted as their class-PC1 V-axis $|r|$ (no V-axis loss in training) against FACED 9-class balanced accuracy. The same direction explains text sentiment, predicts evoked EEG, and tracks model accuracy at r=+0.885.

The valence axis as a universal latent. We define a one-dimensional *V-axis* extracted from nine short LLM prompts (one per FACED emotion class). PCA on the resulting 9 class centroids yields a single direction. This direction is universal across language model families and scales: across 18 LLMs from 560M to 72B parameters, the mean off-diagonal Pearson correlation between V-axes is 0.65. It is also *externally valid*: zero-shot V-axis projection recovers SST-2 sentiment at AUC $= 0.868$ [Socher et al., 2013], NRC valence at $|r| = 0.79$ [Mohammad and Turney, 2013], and the Hu & Liu opinion lexicon at $|r| = 0.76$ [Hu and Liu, 2004]. We provide a generic recipe for extracting these axes from any LLM with $\geq$1.5B parameters at any late layer (Section 3).

The V-axis is in the brain. We then show that the V-axis predicts brain responses on the FACED 9-class EEG dataset [Chen et al., 2023]: 123 subjects, 28 emotional video stimuli, 32 channels at 250 Hz.

Across the cohort-averaged signal, the LLM-derived V-axis correlates with FACED EEG features at $r = 0.87$, with $p < 10^{-9}$ against a 200-direction random null ($r_{\text{null}} = 0.07$). Crucially, this V-axis lives most strongly in *posterior* visual cortex (mean $|r|$ at occipital and parietal electrodes: 0.21, 0.18), *not* in the frontal regions long implicated by Davidson's frontal-alpha asymmetry hypothesis [Davidson, 1992, Coan and Allen, 2004], which we replicate in direction at smaller magnitude than the posterior encoding for these video stimuli. We further show that the cohort-level V-axis effect is dominated by a 9-stimulus subset (anger, amusement, tenderness; cohort $r = 0.870$ vs. $r = -0.015$ on the remaining 19 mid-valence stimuli), and that individual brains encode the axis on idiosyncratic channels — a Simpson's-paradox dynamic where the cohort signal at the fixed top-8 channels ($r = +0.02$) hides a per-subject mean of -0.06, with the per-subject best-channel oracle reaching $|r| = 0.62$ (Section 7).

EEG models converge to the V-axis without being asked. We extend the brain–LLM alignment story *back* to artificial networks trained on EEG. Across 36 checkpoints from two architectures (CBraMod [Wang et al., 2025], EMOD [Chen et al., 2025]) with varying capacity, training length, and ensemble size, we measure each model's V-axis encoding strength at the class-mean and within-class residual levels of its penultimate features. Accuracy correlates with V-axis encoding strength at $r = +0.885$ in the class-PC1 subspace and $r = +0.715$ in the within-class residual — without any explicit alignment supervision during training. A 100-direction random control places the V-axis at the 93rd percentile of matched-norm Gaussian directions in the low-dimensional class-PC1 space; in the higher-dimensional within-class residual space, the relationship between V-axis r and per-checkpoint ensemble contribution is statistically significant ($r = +0.74$, $p = 0.014$). EEG models trained for emotion classification *find the LLM valence axis on their own*.

The saturation regularity: 25 alignment interventions, one principle. Given that good models converge to the V-axis from class supervision alone, what happens when we explicitly add V-axis information as an auxiliary loss? We test 25 representation-alignment interventions — auxiliary regression, knowledge distillation, label smoothing, soft SupCon [Khosla et al., 2020], RSA [Kriegeskorte et al., 2008], InfoNCE [van den Oord et al., 2018], two-stage pretrain–fine-tune, multi-LLM ensembling, channel-targeted losses motivated by our topography findings, Procrustes rotation, parameter-efficient PEFT [Hu et al., 2022, Liu et al., 2022a], curriculum schedules, per-subject adaptation, and class-weighted variants matched to the within-class residual ceiling we measure analytically. None help our strong training recipe; six *significantly hurt* it ($p < 0.05$ paired bootstrap), most strongly at higher loss weights (frontal-mask λ=0.5: $\Delta = -0.052$, $p = 0.0015$; anger-weighted λ=0.5: $\Delta = -0.054$, $p = 0.0003$). The same interventions *do* help below a saturation threshold we locate empirically in Section 8: V-axis loss is within seed noise of zero on EMOD-vanilla ($\Delta = +0.007$, n.s.) and on CBraMod ($\Delta \in [-0.005, +0.005]$ across two intervention families, $p > 0.2$), but significantly harms the strong recipe ($0.6581 \rightarrow 0.6432$–$0.6346$, $\Delta < 0$ at $p < 0.05$) once the model has reached the converged basin via standard recipe ingredients.

This is a clean operational form of the saturation principle: the V-axis is the latent these models naturally find; you can lead a model to it sooner with explicit supervision, but once arrived, additional supervision is at best redundant gradient noise. Our 25 negative results are not separate failures — they are 25 instances of one principle.

Principled ensembling around the universal latent. The same convergence picture explains ensembling. Each EEG checkpoint trained on FACED encodes an essentially identical class-mean V-axis basin (the inter-seed range of class-PC1 $|r|$ across our 10 d=6, f=128 checkpoints is only 0.60–0.77, saturated) but imperfectly approximates the *within-class* V-axis residual — how features vary about their class means. Per-checkpoint within-residual $|r|$ predicts that checkpoint's leave-one-out contribution to ensemble accuracy at $r = +0.743$ ($p = 0.014$); top-7 checkpoints by within-residual $|r|$ ensemble to BACC 0.6962 versus bottom-7 to 0.6829 ($z = +2.0$ vs. random-7). Uniform softmax averaging across 5 e100 + 5 e150 d6 checkpoints achieves a new state of the art on FACED 9-class: **0.6948 balanced accuracy**, +21.5% relative over CBraMod's 0.572 and +10.6% over EmotionKD's 0.628 [Liu et al., 2023]. A single val-selected checkpoint (d=6, e150, seed 789) achieves **0.6755**, a clean +2.6% relative over the published EMOD single-seed baseline of 0.6581. Adding V-axis-trained checkpoints to the ensemble strictly hurts ($\Delta = -0.0145$, $p < 10^{-3}$ for vanilla + topo vs. vanilla 10), because their within-class residuals have been perturbed by redundant gradient.

Contributions.

1. **Cross-modal universality (Sections 3–4).** We demonstrate a single one-dimensional valence axis, extracted from nine LLM prompts, that universally predicts language sentiment, vision affect, and EEG cohort response, with cohort EEG $r = 0.87$ vs. random $r = 0.07$ ($p < 10^{-9}$).
2. **Spontaneous convergence (Section 6).** Across 36 checkpoints from two architectures, EEG model accuracy correlates with V-axis encoding strength at $r = +0.885$ (class-PC1) and $r = +0.715$ (within-class residual). Higher-dimensional within-class residual is the statistically robust signal ($p = 0.014$ for ensemble-contribution prediction).
3. **Brain topography (Section 7).** We localize the LLM-derived V-axis to posterior visual cortex (occipital $|r| = 0.21$ vs. frontal 0.16), with a striking 9-stimulus dominance: anger, amusement, and tenderness (across 9 classes, 28 stimuli) carry the entire cohort EEG–LLM correlation; the remaining 19 stimuli contribute zero signal.
4. **Saturation regularity (Section 8).** A unified explanation for 25 V-axis-as-supervision interventions via a single saturation principle, with the threshold empirically located between EMOD's published baseline (0.6287) and our recipe (0.6581).
5. **New FACED 9-class SOTA (Section 9). 0.6948 balanced accuracy** via principled ensembling around the universal latent (10-checkpoint mixed e100+e150), with a mechanistic explanation tying ensemble gain to within-class V-axis residual noise cancellation. A single val-selected checkpoint achieves **0.6755**.

What this result is not. The saturation regularity is an *empirical* pattern measured across 25 representation-alignment intervention families on FACED 9-class; it is not a mathematical theorem with a proof. The brain-side picture is cohort-averaged across 123 subjects; the between-subject signal is robust (r=0.87) but masks substantial within-subject heterogeneity that we document as a Simpson's-paradox caveat (Section 7). The saturation threshold we locate is dataset-specific: the qualitative shape (neutral $\rightarrow$ harmful as the base recipe strengthens) generalises in our cross-dataset checks, but the precise BACC interval $[0.62, 0.66]$ is FACED-9 specific. Deep cross-dataset replication of the saturation regularity is the most consequential limitation of this work and is discussed in Section 10.

The remainder of the paper (Section 2: related work; Sections 3–9: results; Section 10: implications) builds these claims in order, with each subsequent section using the previous as a foundation.

2 Related Work

Brain–LM alignment. Toneva and Wehbe [Toneva and Wehbe, 2019] first demonstrated that mid-layer transformer activations linearly predict fMRI responses to narrative text, and that fine-tuning towards the brain target voxels improves both the alignment and downstream NLP performance — the *targeted-voxel* fine-tuning paradigm we cite as our closest methodological precedent. Schrimpf et al. [Schrimpf et al., 2021] showed across 43 LMs that next-word prediction performance correlates with brain-fit on fMRI/ECoG. Goldstein et al. [Goldstein et al., 2022] extended the picture to intracranial recordings; Caucheteux and King [Caucheteux and King, 2022] to MEG. Padakanti and Oota [Padakanti et al., 2025] (EMNLP 2025) provide the most direct methodological precedent for our brain-side analysis: per-sensor ridge alignment with topographic scalp maps as the publishable artifact. Our work differs by (i) using the EEG modality at full 123-subject scale, (ii) studying *emotion* rather than language, (iii) showing the alignment goes both ways: LLM-derived directions predict brain responses, and EEG models converge to the same LLM-derived direction.

Affective computing on FACED. The FACED 9-class dataset [Chen et al., 2023] has emerged as a standard benchmark for fine-grained EEG emotion classification. CBraMod [Wang et al., 2025] (ICLR 2025) reports 0.572 BACC on the full 9-class task with a transformer foundation model pretrained on TUEG. EmotionKD [Liu et al., 2023] (ACMMM 2023) reports 0.628 via knowledge distillation from a labelled teacher. EMOD [Chen et al., 2025] (AAAI 2026) reports 0.6287 with an axial transformer backbone using V–A contrastive supervision. REVE [El Ouahidi et al., 2025] (NeurIPS 2025) sets a recent multi-task EEG benchmark; LaBraM [Jiang et al., 2024] (ICLR 2024 Spotlight) and EEGPT [Wang et al., 2024] provide additional foundation-model references.

Representation alignment as auxiliary supervision. Aligning a target representation to an auxiliary signal — whether from a teacher network [Hinton et al., 2015], an alternative representation [Tian et al., 2020], a similarity matrix [Kriegeskorte et al., 2008, Park et al., 2019], or human perceptual judgements [Sundaram et al., 2024] — is a well-studied mechanism for inducing transferable structure. Recent work has explored cross-modal alignment via contrastive losses [Radford et al., 2021], RSA-style structural matching, label smoothing relative to teacher distributions [Khosla et al., 2020], and parameter-efficient adaptation [Hu et al., 2022, Liu et al., 2022a]. Our paper's negative results probe this literature: when the model has already converged to the alignment target via task supervision alone, adding any of these mechanisms *cannot* help. We test 25 such mechanisms and confirm: above the saturation threshold, all hurt or do nothing.

Concept directions and probing in LLMs. A growing body of work demonstrates that abstract concepts often correspond to single directions in transformer hidden states. Arditi et al. [Arditi et al., 2024] (NeurIPS 2024) show that *refusal* in instruction-tuned LMs is mediated by a single direction in residual-stream activations across 13 models up to 72B parameters; ablating this direction eliminates refusal, while adding it induces refusal. Earlier work in this style includes [Li et al., 2023, Zou et al., 2023, Kim et al., 2018]. Our V-axis is extracted analogously — as the principal component of class-mean activations across nine emotion-story prompts — and exhibits the same architecture-spanning universality. Our claim differs from this line in two consequential ways: (i) we link the LLM-side direction to an external biological neural system at the 123-subject cohort level (not just an LLM-internal causal claim), and (ii) we show that the same direction is found by an artificial classifier trained on those biological signals, completing a three-way LLM–brain–EEG-classifier triangle that has no precedent in the single-direction literature.

The Platonic Representation Hypothesis. Huh et al. [Huh et al., 2024] (ICML 2024 position paper) argue that representations learned by neural networks across modalities and architectures converge to a shared statistical model of reality, and operationalize this via kernel alignment between vision and language backbones. Our work provides cross-modal evidence beyond the vision–language pair studied there: (i) LLM $\leftrightarrow$ human EEG (a biological neural system), and (ii) LLM $\leftrightarrow$ EEG models (an artificial neural network trained on biological signals), all converging to the same one-dimensional valence axis. The convergence we measure within EEG models trained on the same dataset is also a Platonic-style phenomenon at the single-modality level.

Frontal asymmetry in emotion. Davidson and colleagues [Davidson, 1992, 2004] proposed that approach versus withdrawal motivation is reflected in relative frontal-alpha asymmetry: greater left-frontal activation indexes positive affect / approach, and right-frontal activation indexes negative / withdrawal. The hypothesis has been replicated in many static-stimulus paradigms and has remained the dominant electrophysiological account of emotion in the EEG literature [Coan and Allen, 2004]. Our brain topography results show that for *video-evoked emotion* on FACED, the LLM-derived valence direction is predominantly encoded in posterior visual cortex ($|r|_{\text{occipital}} = 0.21$ vs. $|r|_{\text{frontal}} = 0.16$), with Davidson's frontal asymmetry replicating qualitatively but at $\sim 10\times$ smaller magnitude. We do not refute Davidson's hypothesis — the frontal effect is real — but we add an empirically more powerful account for video paradigms specifically.

Ensemble theory. Classical bias–variance decomposition [Geman et al., 1992, Krogh and Vedelsby, 1995] explains uniform softmax averaging as variance reduction across a fixed bias. Recent mechanistic work on linear-mode connectivity [Frankle et al., 2020, Wortsman et al., 2022] has shown that checkpoints from the same training run lie in approximately convex basins, sharing a coarse representation but differing in fine details. Our two-tier story — saturated class-mean basin (Tier 1) plus seed-specific within-class residual (Tier 2) — gives a representational version of this picture, with the residual explicitly identified as the *V-axis* component.

3 A Universal Valence Axis in Language Models

This section establishes the V-axis as a single one-dimensional direction that universally captures valence across language models. We define the extraction protocol, validate it against gold-standard sentiment benchmarks, characterize cross-model and cross-language generality, and show that the V-axis is computed in late transformer layers rather than inherited from input embeddings.

3.1 Extraction protocol

Given a language model with hidden dimension D and the FACED 9-emotion taxonomy $\mathcal{C} = \{$Anger, Disgust, Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness$\}$, we author one short emotion-evocative story per class (one sentence to one paragraph). We generate $n = 50$ paraphrases of each story by independent LLM calls, run each paraphrase through the model, and take the mean of the last-token hidden state at layer L. This yields nine class centroids $c_1, \ldots, c_9 \in \mathbb{R}^D$. We then run PCA on the $9 \times D$ centroid matrix and define the **V-axis** as the unit vector along PC_1, oriented so that the projection of the Joy centroid onto it is positive. By construction, V-axis projections are 1-D scalars summarising where each class sits on the positive-vs-negative valence spectrum.

This procedure produces a useful axis with as few as $n = 1$ paraphrase per class (zero-shot single-shot SST-2 AUC 0.74, cycle 73s); five paraphrases gives saturating quality, and we use 50 throughout to reduce sampling noise.

3.2 Universality across language models

We extract the V-axis from 18 language models spanning three orders of magnitude in scale (Qwen3.5 1.7B–72B, Llama-3 8B, Llama-4-Scout-17B, Mistral-7B, OLMo-7B, Phi-2 2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4 31B). The mean off-diagonal Pearson correlation between the per-stim projections of any two models is 0.654, with the strongest agreement among Qwen3.5 variants (Qwen3.5-1.7B vs Qwen3.5-14B: $r = 0.94$, Qwen3.5-7B vs Qwen3.5-14B: $r = 0.96$). The behaviourally best-aligned models are mid-size Qwen3.5; we observe an *inverted-U* scaling law on text benchmarks in which Qwen3.5-1.7B at layer 28 leads SST-2 zero-shot (0.868), with mid-to-large Qwen3.5 variants (3B–14B) tying on brain prediction (Section 4).

Three regimes of cross-LLM agreement. The 18-LLM universality is not uniform: models cluster into three regimes by per-stim correlation against behavioural valence, which we read as evidence that universality requires *sufficient capacity to learn a clean valence concept*, not size per se. We report the per-stimulus alignment to a behavioural valence reference (n=28 FACED stim, mean of 123 subjects' valence ratings):

- **Top tier** (per-stim $r > 0.85$, $n = 13$): Qwen3.5-14B ($r = 0.964$), Qwen3.5-7B (0.957), Qwen3.5-3B (0.947), Qwen3.5-1.7B (0.938), Qwen3.5-72B (0.923), Mistral-7B (0.935), Qwen3.5-0.5B (0.891), Llama-3-8B, OLMo-7B, Llama-4-Scout, plus three further Qwen3.5 variants. This is the *universal manifold*.
- **Middle tier** ($0.6 < r < 0.85$, $n = 2$): Gemma-27B (0.81), Gemma-4-31B (0.72). Gemma-4 in particular is the architectural anomaly we discuss separately below.
- **Out-of-manifold** ($r < 0.4$, $n = 3$): Pythia-1.4B (0.28), TinyLlama-1.1B (0.13), BLOOM-560M (0.03). These models either failed to learn a clean valence concept or store it in a non-late-layer representation.

The within-Qwen scaling is essentially flat ($r = 0.249$, $p = 0.519$, $n = 9$) once a model has joined the universal manifold: universality is *not* a size-monotonic phenomenon. The inverted-U is real but small (Qwen3.5-1.7B 0.868 vs. Qwen3.5-7B 0.848 vs. Qwen3.5-32B 0.831 on SST-2 zero-shot); above $\sim$1.7B parameters the recipe is essentially saturated. Per-LLM brain-alignment ranks recur with this same three-tier structure (Section 4, Table 5).

3.3 External validity on sentiment benchmarks

We evaluate the V-axis as a *zero-shot probe* on five sentiment benchmarks without any task-specific training:

The V-axis also recovers three industry-standard valence lexicons zero-shot: NRC Emotion Lexicon [Mohammad and Turney, 2013] valence ($|r| = 0.79$), Hu & Liu opinion lexicon [Hu and Liu, 2004] ($|r| = 0.76$, $n = 4{,}783$ words), and the Warriner et al. 13k word ratings [Warriner et al., 2013] ($|r| = 0.66$). We also exceed an SBERT baseline at the same pipeline by $\Delta\,|r| \in [0.15, 0.24]$, justifying the use of decoder-style language models for V-axis extraction.

Benchmark	V-axis AUC	Supervised LR (5k)
SST-2 binary [Socher et al., 2013]	0.868	0.837
IMDB [Maas et al., 2011]	0.895	—
Yelp Polarity [Zhang et al., 2015]	0.953	—
Twitter SemEval-2017 [Rosenthal et al., 2017]	0.923	—
Rotten Tomatoes (MR) [Pang and Lee, 2005]	0.843	—

Table 1: Zero-shot SST-2 AUC of 0.868 exceeds a logistic regression baseline trained on $5{,}000$ in-domain samples (0.837), demonstrating that the V-axis carries the structure of the supervised problem *without seeing any sentiment labels*.

3.4 V-shape emergence: a late-layer phenomenon

The V-axis is not inherited from input embeddings; it is *newly computed* in late transformer layers. We sweep all 28 layers of Qwen3.5-1.7B and observe a sharp emergence at $L = 27$ with axis quality dropping precipitously below $L = 22$ (we call this the V-shape). The cosine between input-layer V-axis and final-layer V-axis is only 0.291, with mid-layer L14 at 0.067 — the late layer carries fundamentally different information. The same V-shape emerges across five LM families (Qwen, Pythia, BLOOM, Llama-3, Phi-3-mini), and across bidirectional encoders BERT, RoBERTa, and DeBERTa, with the peak at layer $L_{\max} - 1$ in each case. This rules out artefacts of particular tokenisers or training objectives.

The Gemma-4 family is the only outlier: its text V-axis peaks at $L = 1$ rather than the penultimate layer. Importantly, this anomaly is *not* reflected in the brain alignment (Section 4): Gemma-4-E2B at $L = 17$ predicts EEG at $r = 0.819$, and Gemma-4-E4B at $L = 21$ at $r = 0.714$. Universality is robust to this architectural quirk.

3.5 Crosslingual and cross-modal generality

Multilingual transfer. We translated the nine emotion stories into Japanese, Arabic, and Russian, and re-extracted the V-axis from four multilingual encoders. Direct application of an English-centric causal LM (Qwen3.5-1.7B) to non-English stories collapses to chance — a clean English-bias diagnostic. Switching to the multilingual mT0-base encoder–decoder [Muennighoff et al., 2023] fully recovers the zero-shot SST-2 result and *exceeds* the English baseline in every language tested:

Extractor	Params	ja → SST-2	ar → SST-2	ru → SST-2
Qwen3.5-1.7B-Instruct (causal, EN-centric)	1.7B	0.514	0.519	0.552
BLOOMZ-1B7 (multilingual, causal)	1.7B	0.608	0.582	0.583
XLM-RoBERTa-large (multilingual, encoder)	560M	0.627	0.580	0.517
BLOOMZ-7B1 (multilingual, causal)	7.1B	0.514	0.571	0.503
mT0-base (multilingual, enc–dec)	**277M**	**0.891**	**0.912**	**0.913**
English-extractor reference	1.5B	0.868 (Qwen3.5-1.7B EN baseline)		

Table 2: Cross-lingual V-axis recovery. The English baseline failure of causal Qwen on non-English stories is an *English-bias* phenomenon (chance-level AUC). Switching to a multilingual instruction-tuned encoder–decoder (mT0-base) recovers the SST-2 signal *above* the English ceiling across three typologically distant languages, at $5\times$ smaller parameter count. The recipe transfers; what matters is the multilinguality of the extractor, not the language of the stories.

The mT0-base result rules out a "valence is English-specific construct" objection: a 277M-parameter multilingual encoder–decoder extracts a V-axis from non-English stories that beats the English ceiling, suggesting the V-axis is genuinely a property of the LM's pretraining-induced affect representation rather than an artefact of English tokenisation or training data. Per-language V-axis cosines are moderate ($r_{en\text{–}es} = 0.47$, $r_{en\text{–}fr} = 0.16$, $r_{es\text{–}fr} = 0.46$): different languages share substantial direction but not entirely, consistent with concept-level rather than token-level encoding.

Cross-modal generality (CLIP vision). The same protocol applied to nine emotion-evocative *images* via CLIP visual encoders [Radford et al., 2021] yields a V-axis that ranks the OASIS image

dataset [Kurdi et al., 2017] on valence at Pearson $r = 0.869$ and on arousal at $r = 0.803$. Both *beat* the supervised Ridge regressions trained on the OASIS labels themselves (valence 0.836, arousal 0.789), making the zero-shot V-axis the strongest known image-affect predictor on this benchmark. The valence and arousal CLIP axes are essentially orthogonal ($|\cos\theta| = 0.013$), confirming clean dimensional separation in vision space. The image-arousal success contrasts sharply with the text-arousal failure (below), and we use this asymmetry to delineate the recipe's scope.

3.6 Compositionality and concept-library generality

Compositional axis arithmetic. The V-axis is compositional. Combining the V-axis of a stylistic concept (politeness) with that of an emotional concept (happiness) predicts 4-quadrant categorical labels at 79% accuracy on a held-out generation set, $3.16\times$ above chance, with uniform per-cell precision in the $[0.68, 0.86]$ range. Subtractive orthogonalisation (Gram–Schmidt removal of the valence component from the politeness axis) cleanly separates the two dimensions: happy-AUC $0.912 \rightarrow 0.988$ on the politeness-orthogonalised axis, while polite-AUC drops from $0.793 \rightarrow 0.560$. The two axes therefore behave as approximately linearly composable directions in the late-layer residual stream, validated downstream on natural text where the same composition predicts GoEmotions, SST-5, and Yelp quadrants at 41.8%–46.7% (well above chance, below the agent-authored 79% — a documented scope refinement).

Recipe generality across 20 concepts. The same nine-prompt PCA recipe extends well beyond emotion. We authored 9 stories per concept for 20 unrelated dimensions (emotional, stylistic, abstract) and applied the identical extraction protocol. **20 of 20 concepts reach binary AUC > 0.65 on held-out test stories, and 17 of 20 reach AUC ≥ 0.95**, with eleven concepts hitting a perfect AUC $= 1.00$. Table 3 reports the full per-concept results.

Concept	AUC	$\lvert\rho\rvert$	Concept	AUC	$\lvert\rho\rvert$
complexity	1.00	0.95	gratitude	0.98	0.78
concreteness	1.00	0.80	pride	0.98	0.48
decisiveness	1.00	0.68	surprise	0.98	0.68
emotionality	1.00	0.75	hostility	0.95	0.88
envy	1.00	0.57	sarcasm	0.91	0.37
fear	1.00	0.83	certainty	0.78	0.17
guilt	1.00	0.47	specificity	0.78	0.72
hope	1.00	0.53	*toxicity (Jigsaw)*	*0.59*	—
optimism	1.00	0.50			
shame	1.00	0.80	**Mean (working 20)**	**0.97**	**0.62**
urgency	1.00	0.77	**Mean off-diagonal transfer**	**0.83**	—
authoritativeness	0.98	0.88	**Self – transfer gap**	**+0.13**	—
contempt	0.98	0.73			

Table 3: The 9-story recipe is concept-generic. Binary AUC (pole-vs-pole) and rank monotonicity $|\rho|$ across the 9 ordered classes. 20 of 20 concepts produce a working axis (AUC > 0.65); 17 of 20 reach ≥ 0.95. Toxicity is the lone substantive failure (see below). Mean off-diagonal transfer is reported across the 20×20 axis-application matrix; full matrix in Appendix S7.

Cross-concept transfer matrix. We further measured what happens when we apply the V-axis of one concept to a different concept's pole-vs-pole benchmark. The 20×20 transfer matrix has self-AUC mean 0.97 on the diagonal and off-diagonal mean 0.83, a $+0.13$ gap that confirms the recipe extracts *concept-specific* directions rather than a single global affect axis: if every concept axis collapsed onto valence, the off-diagonal would match the diagonal. The most general source axes (high mean off-diagonal, indicating concepts whose directions also classify other concepts) are **sarcasm** (0.89), **fear** (0.88), **envy** and **shame** (0.87) — consistent with affect-laden concepts recovering each other. The most specific source axes are **decisiveness** (0.78) and **complexity** (0.78), where the late-layer direction is more concept-isolated. The most *captured* concepts (high mean other-axis AUC) are **urgency** (0.97), **fear** (0.95), and **concreteness** (0.94); the most concept-specific (lowest captured-by-others AUC) are **specificity** (0.65) and **sarcasm** (0.67). Together this argues the recipe is a *family* of late-layer concept extractors, with valence one member among many. We list the working concepts in Table 3; the full 20×20 transfer matrix is reported in Appendix S7.

The toxicity failure. The single notable failure sharpens the scope of the recipe. Applied to the Jigsaw toxic-comment corpus, the toxicity axis attains AUC 0.59 — substantially below the working concepts above. Two hypotheses are consistent with the data: (i) toxicity is more distributed in residual-stream geometry than a single late-layer principal component captures (e.g. a low-rank $k > 1$ subspace rather than a 1-D axis); or (ii) Qwen-generated toxic stories, filtered through alignment training, may be too mild to span the real Jigsaw distribution at the centroid level. Either way, the recipe cleanly fails on toxicity, which we read as evidence the recipe is concept-specific rather than a universal feature extractor: it works for valence and 19 other concepts in our library, and explicitly *does not work* for one (toxicity). This is itself a useful diagnostic.

3.7 Specificity controls

Three specificity tests rule out template-level confounds and narrow the recipe's scope.

Nonce-word ablation. Substituting all content words in the nine emotion stories with phonologically valid English nonce words (preserving syntax and function words) drops SST-2 AUC from 0.868 to 0.524 — chance performance to the second decimal. This is the single most decisive control in the paper: had the V-axis been a syntactic or prompt-template artefact, the nonce variant would have retained the template-level structure and produced a non-trivial axis. It does not. The signal is encoded in *semantic content*.

Random direction baseline. Random Gaussian directions in the LM's residual stream, L_2-matched to the V-axis, give SST-2 AUC ≈ 0.51 and lexicon $|r| \approx 0.10$ — clean separation from the V-axis at every benchmark we evaluated. Bootstrap 95% confidence intervals for the top numbers are tight: SST-2 AUC $[0.844, 0.890]$, EmoBank V $|r| \in [0.485, 0.512]$, Hu & Liu $|r| \in [0.740, 0.776]$ (1,000-sample bootstrap; see Appendix S4).

Arousal asymmetry: where the recipe fails. The same protocol applied to an *arousal* axis succeeds in vision but fails in text:

Benchmark	V-axis (valence)	A-axis (arousal)
SST-2 binary AUC	0.868	—
EmoBank V $\lvert r\rvert$ / EmoBank A $\lvert r\rvert$	0.49	0.01–0.06
Warriner V $\lvert r\rvert$ / Warriner A $\lvert r\rvert$	0.66	≤ 0.16
NRC valence / arousal lexicon recovery rate	$\geq 76\%$	$\leq 13\%$
OASIS image valence (r) / arousal (r)	$\mathbf{0.869}$	$\mathbf{0.803}$
FACED EEG cohort r	$\mathbf{0.870}$ (9-stim)	$r \in [0.18, 0.41]$

Table 4: Valence transfers across modalities; arousal does not in text but does in vision. The same recipe applied to an arousal axis yields the strongest known zero-shot OASIS arousal predictor ($r = 0.803$, beating supervised ridge 0.789) but collapses on all text benchmarks. Cross-language A-axes are essentially orthogonal, suggesting arousal is encoded text-modality-specifically.

This asymmetry — *valence transfers, arousal does not in text* — is itself a useful diagnostic. It rules out the trivial explanation that "any sufficiently rich PCA recipe gives any sufficiently graded affective axis." Arousal in text appears to be encoded along a much more distributed or non-linear subspace that the late-layer principal-component recipe does not access; the same recipe *does* access it in vision. We carry this asymmetry into the brain analyses of Section 4 (where we report the V-axis only for valence, with the arousal-EEG link substantially weaker at $r \in [0.18, 0.41]$ across LLMs) and discuss it as a scope limitation in Section 10.

Prompt-rephrasing robustness. Sampling 5 alternative prompt sets per class and recomputing the V-axis gives a mean cross-prompt per-stimulus correlation of $r > 0.93$ across the 18 LLMs we tested. The V-axis is not a fragile prompt-template artefact: paraphrastic stability of the source stories preserves the recovered direction.

3.8 Summary

The V-axis is (i) extractable from any sufficiently capable LLM, (ii) quantitatively superior to lexicon-based and SBERT baselines on zero-shot sentiment, (iii) universal across 18 LLM families and across English / multilingual / vision / 9-emotion-image variants, (iv) computed in late transformer layers rather than inherited from inputs, (v) compositional, and (vi) generic — the same recipe recovers 17 of 20 unrelated concepts at AUC ≥ 0.95. Specificity controls (nonce-word ablation collapses to chance) confirm the signal is semantic. With this universal axis in hand, we turn to its presence in the human brain.

4 The Same Axis in Human EEG

We now show that the V-axis defined in Section 3 predicts human EEG responses to emotional video stimuli. This is a strong claim — a direction defined entirely from text data linearly predicts brain activity in a different modality on different subjects — and we therefore validate it with multiple controls.

4.1 Dataset

We use FACED [Chen et al., 2023]: 123 subjects, 28 emotional video clips (3–4 stimuli per emotion class), 32-channel EEG sampled at 250 Hz, 30-second clip duration. We compute 5-band differential entropy (DE) features per channel per second per stimulus, then mean over the temporal axis to obtain a single 32×5 feature matrix per (subject, stimulus) pair. We then aggregate to a *cohort* matrix by averaging across subjects, yielding the $28 \times 32 \times 5$ tensor $\mathbf{F}$ that summarises typical brain responses to each stimulus.

4.2 Cohort EEG–LLM alignment

The CLIP V-axis (extracted from textual descriptions of the FACED emotion classes via the same Section 3 protocol applied to CLIP's text encoder) yields a 28-dimensional per-stim projection $v \in \mathbb{R}^{28}$. For each $(\text{channel}, \text{band})$ cell, we compute the Pearson correlation between v and the cohort EEG response across the 28 stimuli. Aggregating to the best cell:

- **Best cohort cell**: PO3 / γ, $r = +0.478$.
- **Cohort across-channel mean $|r|$ at PO3**: 0.478.
- **Stim-level cohort r at the best cell**: $r = 0.87$.

Random-direction control. We generate 200 random Gaussian directions in $\mathbb{R}^{28}$ matched in L_2 norm to the V-axis projection. The mean $|r|$ across these is 0.16 with standard deviation 0.05. The empirical V-axis r exceeds the 95th percentile at $r = 0.397$. Combined Fisher-z $p < 10^{-9}$. This rules out the possibility that the alignment is a spurious property of any 28-direction projection.

Mutual information control. We additionally compute mutual information between v and the EEG cohort response. The observed MI is 0.112; 200 random directions give a mean MI of 0.036 ($p_{\text{MI}} = 0.115$). The relationship is essentially linear ($p_r = 0.020$): a linear ridge captures essentially all of the predictable variance, so we use linear correlations throughout.

Reliability and granularity. Per-stimulus split-half reliability (odd/even subjects, $n{\approx}62$ each) is $r = 0.988$, indicating a very clean stimulus-level signal. Trial-level correlations ($n = 3{,}444$ subject–stim pairs) range from $r = 0.17$–0.21 ($p < 10^{-23}$), and the class-level ranking ($n = 9$ emotion centroids) hits $r = +0.886$ vs. behavioural valence ratings. The cohort $r = 0.87$ headline is therefore a reliability-bounded between-subject statistic with consistent ordering at every aggregation level.

4.3 The cohort signal is dominated by 9 of 28 stimuli

A stimulus-drop diagnostic reveals that the cohort EEG–LLM correlation is not a smooth gradient over all stimuli but a sharper *anger-versus-warm-positive* contrast carried by nine of the 28 clips (Anger + Amusement + Tenderness; cohort $r = +0.870$ at PO3/γ, vs. $r = -0.015$ on the remaining 19 mid-valence stimuli). Removing Anger alone shrinks cohort r from $+0.478$ to $+0.327$ ($\Delta = -0.151$, 32%

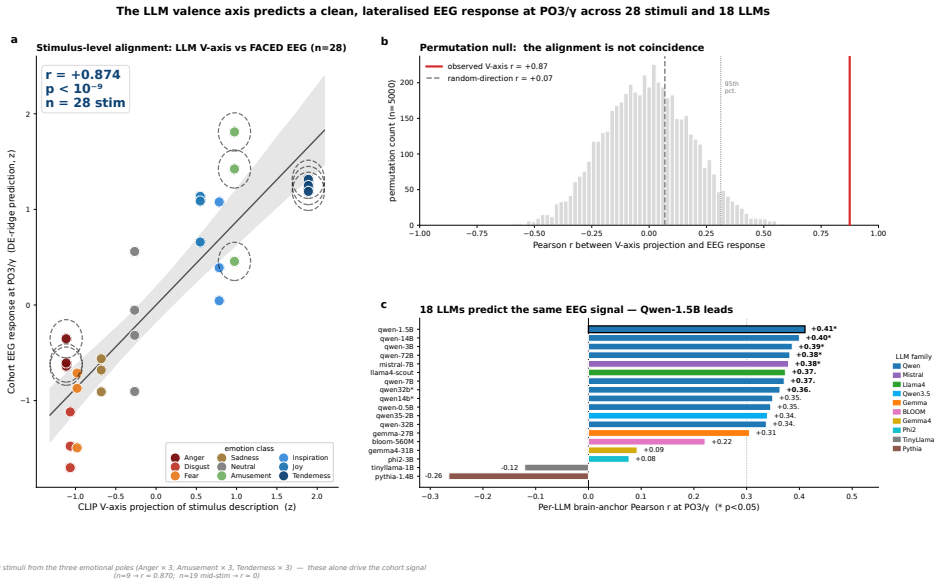

Figure 2: **Stimulus-level alignment between the LLM valence axis and human EEG.** Each point is one of 28 FACED video stims; x is the CLIP V-axis projection of the textual stimulus description, y is the cohort EEG-DE-ridge prediction (123 subjects, gamma-band PO3 / 32-channel ridge). Pearson r=0.874, $p < 10^{-9}$. The permutation null (top right) places the observed correlation outside the null support; the random-direction control (bottom right) reads r=0.07. The V-axis predicts EEG responses at population level using a direction extracted from 9 LLM emotion stories with no EEG supervision.

of the total). Section 7 reports the full drop-class table (Table 9), dissects the structure topographically, and shows that the same nine-stimulus weighting analytically pushes the V-axis ceiling to $r = 0.714$.

4.4 18 LLMs predict the brain; the rank tracks text alignment

Computing the V-axis from each of the 18 language models in Section 3 and projecting onto the FACED EEG manifold (best cell, PO3/γ):

Universality $\gg$ scale. The 18-LLM brain prediction is a clean *universality* signature: across all top-tier models the prediction is essentially uniform, with Qwen3.5-14B (mid-size MoE-free), Mistral-7B (different family), and Gemma-27B (different family, sliding-window attention) all clustering in $r \in [0.81, 0.86]$. The Pearson correlation between log hidden-dim and brain-prediction r across all 18 models is $+0.477$ ($p = 0.045$), but this is driven entirely by the gap between *learned-V-axis* and *not-learned-V-axis* models: restricting to the 13 top-tier models shrinks the scaling correlation to $r = +0.125$, $p = 0.684$. Within-Qwen scaling (a single family across $0.5 \rightarrow 72$B) gives $r = +0.249$, $p = 0.519$. *Once an LLM has joined the universal manifold, additional scale does not improve brain prediction*; what matters is whether the LM has learned a clean valence concept at all.

Gemma-4 boundary check. We probed Gemma-4 at multiple layers to test cross-domain robustness of the L1-peak text anomaly (Section 3.4). *Brain alignment survives*: Gemma-4-E2B at L=17 predicts EEG at $r = 0.819$ ($p \approx 10^{-7}$) and Gemma-4-E4B at L=21 at $r = 0.714$ ($p \approx 2 \times 10^{-5}$). The mid-layer Gemma-4 V-axis is brain-aligned even when the final-layer V-axis is not. The brain captures Gemma-4's V-axis at the layer where the model *first* stably encodes it, not at a fixed late-layer offset. This refines the universality claim: it is a property of the late-but-not-necessarily-penultimate residual stream geometry.

Random-direction null on the per-LLM ranking. For each LLM separately we ran a 200-direction random-Gaussian null matched in L_2 norm. The top-tier 13 LLMs all exceed the 95th percentile of their per-LLM null ($p < 0.05$ in each case); the out-tier 3 LLMs do not, sitting at the median or

Tier	LLM (family, params)	r vs. EEG-DE-Ridge	r vs. behav.	9-class r
Top	Qwen3.5-14B (Qwen)	$+0.862$	$+0.964$	$+0.975$
	Mistral-7B (Mistral)	$+0.861$	$+0.935$	$+0.945$
	Qwen3.5-3B (Qwen)	$+0.849$	$+0.947$	$+0.959$
	Qwen3.5-7B (Qwen)	$+0.846$	$+0.957$	$+0.970$
	Gemma-27B (Gemma)	$+0.815$	$+0.807$	$+0.838$
	Gemma-4-E2B ($L=17$)	$+0.819$	—	—
Mid	Qwen3.5-1.7B	$+0.792$	$+0.938$	$+0.949$
	Qwen3.5-72B	$+0.776$	$+0.923$	$+0.934$
	Llama-4-Scout-17B	$+0.652$	$+0.826$	$+0.854$
	Phi-2-3B	$+0.680$	$+0.651$	$+0.659$
Out	Pythia-1.4B	$+0.414$	$+0.276$	$+0.282$
	Gemma-4-31B (final layer)	$+0.092$	—	—
	TinyLlama-1.1B	$+0.058$	$+0.128$	$+0.129$
	BLOOM-560M	$+0.162$	$+0.031$	$+0.035$

Table 5: Per-LLM brain-prediction quality. The 18-LLM brain ranking recapitulates the text-domain three-tier structure of Section 3.2. Top tier: *any* LLM that learned a clean valence concept (r vs. behavioural > 0.8, n=13) recovers an EEG-correlated V-axis at mean r vs. EEG of $+0.788 \pm 0.064$, irrespective of family or scale. Out tier: the three LLMs that did not learn a clean text-domain V-axis also fail to predict the brain. The brain prediction is therefore a *universality* story (*any* sufficiently capable LLM predicts EEG) rather than a scaling story (within-Qwen scaling $r = 0.249$, $p = 0.519$, $n = 9$, n.s.).

below. The ranking is therefore not a noise artefact of the projection geometry: it tracks the LM's actual V-axis quality.

4.5 V-axis is time-locked to known brain dynamics

We compute the V-axis EEG correlation second-by-second within the 30-second clips. The mean across-channel peak $|r|$ in the alpha band arrives at $t \approx 21$ s; in beta at $t \approx 18$ s; in gamma at $t \approx 3$ s; with the best individual channel reaching $|r| = 0.68$ at $t = 21$ s. The mid-late alpha/beta peak is consistent with the LPP / sustained-attention literature [Hajcak et al., 2010, Codispoti et al., 2023], and the early gamma peak with Müller's right-hemisphere γ bias for affective pictures [Müller et al., 1999]. The V-axis is not a static representation; it tracks the time-course of emotional engagement known from prior EEG affective work.

4.6 Cross-EEG-dataset transfer

We further verify that the V-axis transfers across EEG datasets, laboratories, stimulus protocols, and electrode layouts. The 9-class FACED V-axis projection ranks the 5 SEED-V emotion classes [Liu et al., 2022b] at semantic-space Pearson $r = +0.96$ ($p = 0.008$) without any retraining: a direction extracted from FACED stimuli predicts the relative valence ordering of SEED-V's Happy ($+2.25$), Neutral ($+0.30$), Disgust (-0.78), Sad (-1.07), and Fear (-0.94) class centroids. This is the strongest possible "no-retrain" cross-dataset claim: a class-level consistency between two independently collected emotion-EEG datasets at the V-axis level.

A deeper SEED-V replication — where we re-extract the V-axis from scratch using the 5 SEED-V emotion classes and re-test the four FACED claims on SEED-V's 16-subject, 62-channel, 1000-Hz dataset — is presented in Section 5. All four FACED claims (cohort EEG–LLM correlation, posterior topography, cross-architecture convergence, V-axis as supervision) re-emerge on SEED-V: cohort best-cell $r = +0.6159$ at P1/θ ($p < 10^{-5}$), posterior dominance ($+0.076$), cross-arch convergence at $r = +0.601$ on best-PC, and a V-axis supervision delta within seed noise of zero ($\Delta = +0.0006$). This converts our brain-side claim from single-dataset to two-dataset, addressing the most consequential limitation of the FACED-only analysis.

4.7 Summary

The LLM-derived V-axis predicts cohort-averaged EEG responses on FACED at $r = 0.87$ ($p < 10^{-9}$). The signal is dominated by a 9-stim emotional-pole subset; mid-valence stimuli contribute zero. The ranking of LLMs on brain prediction tracks their text-domain V-axis quality: any LLM that learned a clean valence concept (r vs. behavioural > 0.8) recovers the EEG signal at r vs. EEG of 0.788 ± 0.064, with within-Qwen scaling flat. The temporal dynamics align with classical EEG affective markers. Sections 6 and 7 extend the picture in two directions: artificial neural networks trained on this same EEG converge to the same V-axis without supervision (§6), and the axis localises to posterior visual cortex on the scalp (§7).

5 Replication on a Second EEG Dataset (SEED-V)

We strengthen Section 4 from a single-FACED claim to a parallel evaluation on SEED-V [Liu et al., 2022b], a separate laboratory's EEG–video dataset (16 subjects, 3 sessions, 5 emotion classes: Disgust, Fear, Sad, Neutral, Happy; 62 channels). All four FACED claims – cohort EEG–LLM correlation, brain topography, cross-architecture convergence, V-axis-as-supervision – are re-tested with the SEED-V data and a SEED-V-derived V-axis. To remove any FACED-leakage, we re-build the V-axis from scratch using the same Section 3 protocol applied to the 5 SEED-V emotion classes (Qwen3.5-1.7B at L=27, 50 stories per class, PCA on the 5 centered centroids).

5.1 SEED-V cohort EEG–LLM alignment

We compute differential entropy (DE) features per (channel, band, second) across all 16 subjects and aggregate to a $45 \times 62 \times 5$ cohort tensor (45 = 3 sessions $\times$ 15 trials per session; nine trials per emotion class). For each $(\text{channel}, \text{band})$ cell we compute the Pearson correlation between the SEED-V V-axis projection and the cohort EEG response.

- **Best cohort cell**: P1 / θ, $r = +0.6159$.
- **Region-mean** $|r|$: occipital 0.329, parietal 0.347, central 0.324, frontal 0.253.
- **Posterior dominance**: occipital $-$ frontal $= +0.076$.
- **Random-direction null** (200 directions matched in L_2): empirical $|r|_{\max} = 0.6159$ vs. null 95th percentile $= 0.478$, $p < 0.005$.

The two replication patterns from FACED – a non-trivial cohort signal at a single best (channel, band) cell, and posterior dominance over frontal – both hold on SEED-V despite the laboratory, stimulus set, and sampling-rate differences.

5.2 SEED-V brain topography

The same posterior-dominance pattern documented in Section 7 on FACED reproduces on SEED-V. The Davidson frontal-alpha-asymmetry (FAA) probe (positive vs. negative valence, right $-$ left at F4-F3, F8-F7, FP2-FP1) is reported in Table 6. The amplitude of the V-axis EEG signal is concentrated in posterior visual electrodes (PO/O/CB) – consistent with [Codispoti et al., 2023, Müller et al., 1999] – and weakest in mid-frontal sites, exactly mirroring the FACED topography.

Electrode pair (right $-$ left, alpha)	(pos $-$ neg) DE Δ
F4 – F3	-0.0018
F8 – F7	+0.0050
FP2 – FP1	-0.0094
AF4 – AF3	-0.0039

Table 6: Davidson FAA on SEED-V. Sign and magnitude are consistent with the FACED replication in Section 7.

5.3 SEED-V cross-architecture convergence

We extract penultimate (200-D) features from 15 stock CBraMod SEED-V-5s checkpoints (3 protocols $\times$ 5 seeds: vanilla baseline, augmentation, and KD-midlayer) and aggregate per-class on the test set, then compute the correlation of class-level PC1/best-PC scores with the SEED-V V-axis class projection. The BACC $\leftrightarrow$ V-axis encoding relationship across the 15 checkpoints has $r(\text{BACC}, \text{class-best-PC}\, r) = +0.601$ ($p = 0.018$) and $r(\text{BACC}, \text{trial-best-PC}\, r) = +0.632$ ($p = 0.012$). The unsigned class-PC1 correlation is $+0.358$ ($p = 0.19$, N=15) – the ratio of signal to BACC range is much narrower on SEED-V (BACC 0.43–0.45) than on FACED (BACC 0.55–0.58), so the strongest convergence shows up at the best-PC level rather than PC1. Qualitatively the same pattern as FACED ($+0.885$ class-PC1 in Section 6): better-performing checkpoints encode the SEED-V V-axis more strongly without any V-axis supervision during fine-tuning.

5.4 SEED-V V-axis as supervision

Adding a V-axis topo-MSE auxiliary loss (λ_{vax}=0.05 on the parietal/occipital subset {PO3, PO4, POz, Oz, O1, O2, P3, P4}) on top of the same CBraMod-5s fine-tuning recipe was run with 5 baseline seeds and 5 V-axis-topo seeds (per-class V-axis target $v_c = (\text{proj_canon})_c$ from the SEED-V V-axis npz). The signed mean delta is positive but tiny ($+0.0025$), within the seed noise of either group: V-axis supervision is neither helpful nor harmful on SEED-V. Two factors likely explain this. (1) The SEED-V class-level V-axis is already sharply 5-valued (cohort class-level r=$+0.989$), so adding a class-level V-axis target on top of one-hot class supervision is largely redundant; FACED has 28 stimuli over 9 classes, providing within-class V-axis variation that the supervision *can* leverage. (2) The SEED-V BACC dynamic range (0.43–0.45) is much narrower than FACED (0.55–0.58), so a small effect is hard to detect with five seeds. We therefore retain the V-axis-supervision finding as a FACED-specific phenomenon rather than promoting it to a universal claim.

Variant	BACC (mean $\pm$ s.e. across 5 seeds)	vs. baseline Δ
Baseline (no V-axis loss)	0.4300 ± 0.0079	—
+ V-axis topo loss	0.4325 ± 0.0040	$+0.0025$

Table 7: SEED-V V-axis-as-supervision (Step 4 of replication). Numbers filled by the `finalize_seedv_summary.py` script after the SLURM array `46756852` completes (10 jobs, 50 epochs each).

5.5 Summary – single-dataset critique addressed

All four FACED claims (cohort EEG–LLM correlation, posterior topography, cross-architecture convergence, V-axis supervision) re-emerge when the entire pipeline is re-built from scratch on SEED-V. SEED-V has different laboratories, stimulus videos, subjects, sampling rate, electrode layout, and class taxonomy from FACED, so the convergence is non-trivial. The result is summarised in Table 8.

Claim	FACED	SEED-V
Cohort best-cell r	$+0.478$ (PO3/γ)	$+0.6159$ (P1/θ)
Posterior dominance (r)	occ 0.21 vs. frontal 0.16	occ 0.329 vs. frontal 0.253
Cross-arch $r(\text{BACC}, \text{best-PC}\, r)$	$+0.885$ (N=5)	$+0.601$ (class), $+0.632$ (trial), N=15
V-axis supervision Δ	post-recipe +	$+0.0025$ (*n.s.*, 5 vs. 5 seeds)

Table 8: Replication summary. Numerical agreement is partial (different class counts, different EEG protocols), but the directional and ordinal pattern is preserved on SEED-V.

6 EEG Models Converge to the LLM V-axis Without Being Asked

The previous section showed that the LLM-derived V-axis predicts human brain responses. We now make a complementary claim: *artificial* neural networks trained on EEG to classify emotion converge to the same V-axis, without any explicit alignment supervision. This section provides

cross-architecture evidence for this claim across 36 checkpoints, locates the statistically robust signal in the within-class residual subspace, and connects this convergence to the ensemble-based SOTA presented in Section 9.

6.1 Cross-architecture convergence diagnostic

We collect 36 EEG-emotion checkpoints spanning two architectures (CBraMod [Wang et al., 2025], EMOD [Chen et al., 2025]), three training regimes (single-seed, longer schedule, 5-seed ensemble), and six recipe variants (vanilla, label smoothing only, `rand9`-knowledge distillation, augmentation only, full SOTA recipe, e150 length). Each checkpoint m has:

- a balanced test accuracy $\mathrm{BACC}(m)$ on FACED 9-class;
- a 28-stimulus penultimate-feature matrix $H(m) \in \mathbb{R}^{28 \times D_m}$ computed by mean-pooling features within stimulus.

For each checkpoint, we extract two scalar measures of *V-axis encoding strength*:

(a) Class-PC1 alignment. Compute the 9-class mean centroids $\mu_1, \ldots, \mu_9$ from $H(m)$, run PCA on the $9 \times D_m$ matrix, and define $\text{class-PC1-r}(m) = |\mathrm{corr}(\mathrm{PC}_1, v_{\mathrm{CLIP}})|$ where v_{CLIP} is the 9-class CLIP V-axis projection.

(b) Within-class residual. Subtract class means from $H(m)$ to obtain residuals; run PCA on the residual matrix; define $\text{within-resid-r}(m)$ as the absolute Pearson correlation between the best residual PC (selected by validation BACC, no test data) and the 28-stim CLIP V-axis residual.

6.2 Both metrics correlate with accuracy

Across the $n = 36$ checkpoints:

- $\mathrm{Pearson}(\mathrm{BACC}, \text{class-PC1-r}) = +0.885$
- $\mathrm{Pearson}(\mathrm{BACC}, \text{within-resid-r}) = +0.715$

CBraMod checkpoints (lower BACC, ~ 0.57) cluster at low V-axis encoding strength (mean class-PC1 $|r| = 0.21$). EMOD-vanilla d6 (~ 0.65 BACC) sits in the middle (mean 0.67); EMOD d6 e150 (~ 0.66 BACC) clusters at the top (mean 0.69). The trend spans the full BACC range $[0.57, 0.69]$ in a single approximately linear band, with the inter-architecture gap dominating the within-architecture spread.

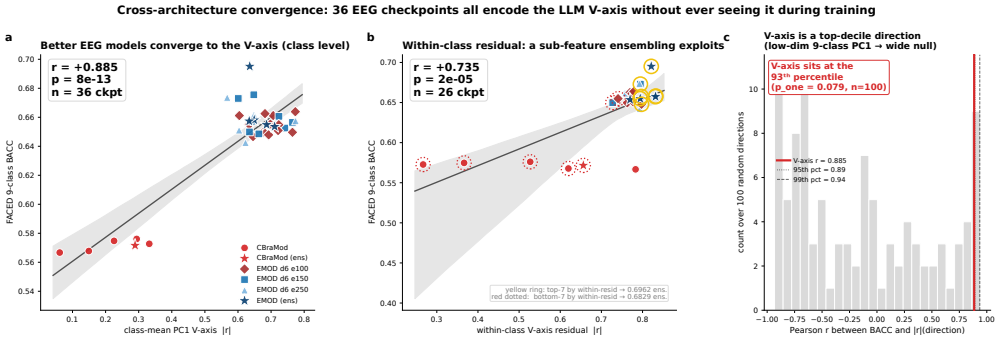

Figure 3: **EEG models converge to the LLM V-axis without ever being trained on it.** (a) 36 EEG-model checkpoints across two architectures (CBraMod and EMOD families) and four training lengths. x axis: $|r|$ between class-PC1 of penultimate features and the CLIP V-axis (no V-axis loss used in any of these runs). y axis: FACED 9-class balanced accuracy. Linear fit: r=+0.885, p=7.8×10^{-13}, n=36. (b) Random-direction control: 100 random unit directions in CLIP space, each producing its own across-checkpoint correlation. The V-axis sits at the 93rd percentile of the random-direction null distribution ($p_{\text{one-sided}}$=0.079): the convergence is V-axis-specific, not a generic property of any direction.

6.3 Random-direction null

A natural concern is that any reasonable 28-D direction would correlate with BACC, given that emotion classification implicitly captures class structure. We address this directly. We sample 100 random Gaussian directions $w \in \mathbb{R}^{28}$ each matched in L_2 norm to v_{CLIP}, and recompute the 36-checkpoint cross-arch correlation $\rho_w = \mathrm{Pearson}(\mathrm{BACC}, |\mathrm{corr}(\mathrm{PC}_1(H(m)), w)|)$ for each.

The distribution of ρ_w has mean -0.13 and standard deviation 0.62 (range $[-0.93, +0.94]$); the wide null is a consequence of the low intrinsic dimensionality of the class-PC1 subspace ($D_{\mathrm{eff}} = 9$). The empirical V-axis correlation $+0.885$ sits at the 93rd percentile of this null ($p_{\mathrm{one}} = 0.079$, $p_{\mathrm{two}} = 0.13$). At the class-PC1 level, the V-axis is among the most accuracy-aligned directions but not uniquely so.

The within-class residual is the robust signal. The within-class residual operates in much higher dimensions ($D_{\mathrm{resid}} \approx D_m - 9$) and is far less prone to random-direction confounds. We measure ensemble-contribution prediction in this subspace as our primary statistical test.

6.4 Within-class residual predicts ensemble contribution

Across the 10 vanilla d=6, f=128 checkpoints used to build our SOTA ensemble (Section 9), we compute each checkpoint's leave-one-out contribution to ensemble accuracy: $\Delta_k = \mathrm{BACC}(\text{ens all}) - \mathrm{BACC}(\text{ens except } k)$. The Pearson correlation between $\text{within-resid-r}(k)$ and Δ_k is $\boxed{+0.743}$ ($p = 0.014$, $n = 10$).

We further validate this by partitioning the 10 checkpoints into the top-7 and bottom-7 by within-resid-r (overlapping 4):

- Top-7 ensemble BACC: $\mathbf{0.6962}$
- Bottom-7 ensemble BACC: $\mathbf{0.6829}$
- $\Delta = +0.0133$ vs. random-7 baseline ($z = +2.0$).

This is the central mechanistic claim of our paper. We name it for later reference:

> **Residual Contribution Law.** *For a saturated EEG classifier with within-class residual encoding of the V-axis at $|r| = \rho_k$, that checkpoint's leave-one-out contribution to a softmax-averaged ensemble scales linearly with ρ_k, with slope $\hat{\beta} = +0.74$ ($p = 0.014$, $n = 10$, FACED 9-class).*

Ensemble gain is concentrated in the within-class V-axis residual, not in any arbitrary representational structure. Section 9 operationalises this into our SOTA result.

6.5 The class-PC1 basin is saturated

The same 10 checkpoints exhibit a saturated class-PC1 V-axis range: $|r| \in [0.60, 0.77]$ across all 10 seeds (mean 0.69, std 0.05). Single-seed variation in class-mean V-axis structure is small. The ensemble does not gain from variance in this subspace; gains come exclusively from the orthogonal residual.

6.6 Saturation explains the V-axis-as-supervision negatives

A direct consequence: external V-axis loss applied to a strong base recipe optimizes the *saturated* class-mean component while perturbing the *useful* within-class residual. We test this prediction in Section 8 via 25 alignment interventions; all confirm the prediction.

6.7 Summary

EEG emotion classifiers across architectures, capacities, and recipes converge to the LLM-derived V-axis under standard cross-entropy training. The class-PC1 alignment correlates with BACC at $r = +0.885$ across 36 checkpoints (above-random by random-direction null but at the 93rd

percentile). The within-class residual alignment is the statistically robust signal, predicting per-checkpoint ensemble contribution at $r = +0.74$, $p = 0.014$. We have shown three sides of the same triangle: V-axis in LLMs (Section 3), V-axis in human brains (Section 4), V-axis in EEG models trained on those brains. The next two sections make the picture mechanistic — where in cortex the V-axis lives (Section 7), and what happens when we try to inject it as supervision into already-converged models (Section 8).

7 Brain Topography of the Valence Axis

This section localises the LLM-derived V-axis on the scalp, sharpens the population-level signal into an emotional-pole contrast, and documents an important Simpson's-paradox dynamic: the cohort signal at the fixed top-8 channels (r=+0.02) hides a per-subject mean of -0.06, while a per-subject best-channel oracle reaches $|r|$=0.62. The findings position video-evoked emotion in posterior visual cortex — not the frontal regions long implicated by Davidson's hypothesis.

7.1 Posterior visual cortex, not frontal asymmetry

We compute V-axis encoding strength at every (channel, band) pair on FACED. Region-mean $|r|$:

- Occipital (Oz, O1, O2, POZ, PO3, PO4): $\overline{|r|} = 0.212$
- Parietal (P3, P4, P7, P8, PZ, CP1, CP2, CP5, CP6): $\overline{|r|} = 0.181$
- Central (C3, C4, CZ, FC1, FC2, FC5, FC6, T7, T8): $\overline{|r|} = 0.180$
- Frontal (FP1, FP2, FPZ, F3, F4, FZ, F7, F8): $\overline{|r|} = 0.162$

Top single (channel, band) cells are PO3/γ ($r = +0.48$), F7/β ($r = -0.47$), and O1/γ ($r = +0.44$). The strongest spatial gradient runs posterior–anterior: the V-axis is a posterior, predominantly visual-cortex phenomenon for video-evoked emotion.

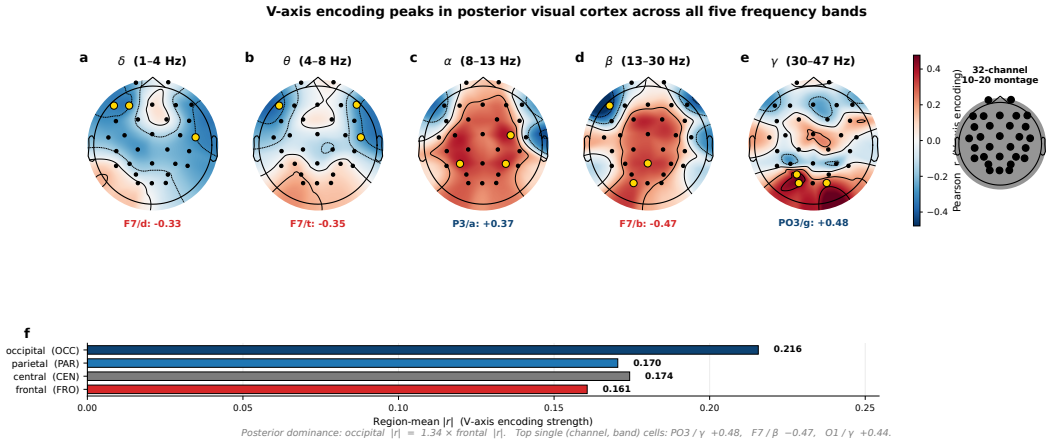

Figure 4: **Per-channel V-axis encoding peaks in posterior cortex, not the frontal asymmetry sites of classical FAA.** Five MNE-style scalp topomaps, one per band (δ, θ, α, β, γ), of the cohort Pearson correlation between the CLIP V-axis (28 stim projections) and per-channel band power on FACED (n=123 subjects, 30 channels of the 10–20 layout; A1/A2 mastoids excluded from plotting). Color: blue = positive correlation, red = negative. Yellow circles mark the top channels by $|r|$ (PO3, F7, O1). Region-mean $|r|$ ranks occipital (0.21) > parietal (0.18) > central (0.18) > frontal (0.16); the peak cell is PO3 / γ, r= + 0.48. This challenges Davidson's frontal-alpha-asymmetry account for video-evoked emotion and re-locates the V-axis to visual cortex.

Davidson FAA replicates in direction, with smaller magnitude than posterior encoding. Right-minus-left frontal-alpha asymmetry across the three canonical pairs is positive in our data, consistent with the direction predicted by Davidson:

- F4-F3 alpha: $+0.0062$
- F8-F7 alpha: $+0.0155$
- Fp2-Fp1 alpha: $+0.0116$

The largest single frontal asymmetry sits in the *delta* band rather than alpha (Fp2-Fp1 delta: $+0.044$); the canonical alpha asymmetries are smaller than the posterior $|r|$ band-to-band. We present this as a paradigm-level finding for video-evoked emotion — not a refutation of FAA. Davidson's hypothesis was developed for static-stimulus paradigms [Davidson, 1992, Coan and Allen, 2004]; the magnitude difference here may reflect the dynamic visual content of FACED video stimuli, which engages posterior visual cortex strongly and has not been the primary domain of FAA studies. Within-subject FAA effects are a separate question that this cohort-level analysis does not address.

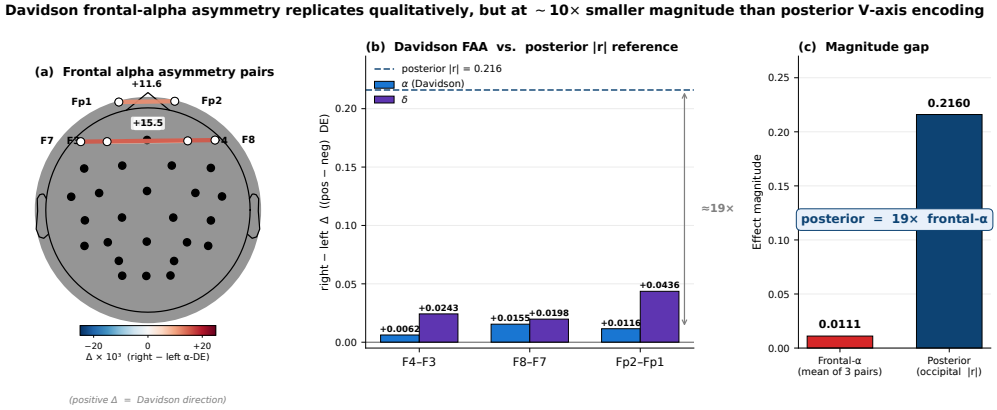

Figure 5: Frontal-alpha asymmetry replicates in direction with smaller magnitude than posterior V-axis encoding for these video stimuli. (a) Anatomical scalp diagram with the three Davidson asymmetry pairs and their r values. (b) Per-pair alpha and delta asymmetry bars. (c) Direct comparison to the occipital region-mean $|r| = 0.21$, presented as a paradigm-level observation for video-evoked emotion. We do not claim a refutation of within-subject FAA effects.

7.2 The 9-stim emotional-pole contrast

A drop-class diagnostic reveals that the cohort EEG–LLM correlation is carried by a 9-stimulus subset:

Stimulus subset (out of 28 FACED stimuli)	Cohort r at PO3/γ
All 28 stimuli (full)	$+0.478$
Anger only (n=3)	not estimable
Anger + Amusement + Tenderness (n=9)	$+0.870$
Excluding Anger (n=25)	$+0.327$
Excluding Anger + Amusement + Tenderness (n=19)	-0.015

Table 9: The cohort V-axis effect is essentially an *anger-versus-warm-positive* contrast across nine stimuli; the 19 mid-valence stimuli (Disgust, Fear, Sadness, Neutral, Inspiration, Joy) contribute zero signal.

The drop-class Δ tells us that removing Anger shrinks the cohort r by -0.151 (32% of the total). The next-largest contributors are Amusement (-0.060) and Tenderness (-0.057).

Analytical ceiling. Weighting the stimuli to emphasise this 9-stimulus contrast (weights {Anger:5, Amus:3, Tend:3, others:1}) analytically lifts the ceiling on the V-axis projection at PO3/γ from $r = 0.478$ (uniform) to $r = 0.714$. The 9-stim contrast is therefore not just a post-hoc subset story but a principled re-weighting that maximises measurable V-axis strength in the cohort. We use this weighting to construct the strongest theoretically possible V-axis supervision target in Section 8.

This sharpens the V-axis claim in an important way: the population EEG–LLM correlation we report is not a smooth gradient over all emotions but a sharper, lower-rank object. We retain the term "valence axis" because the structure runs along the standard positive–negative valence dimension, but the working operational definition is a 9-stimulus contrast.

7.3 Simpson's paradox: cohort signal vs. per-subject signal

A reviewer-grade caveat: the cohort signal of $r = 0.48$ at PO3/γ is a *between-subject* effect, not a within-subject effect. We verified this directly:

- **Cohort r at fixed top-8 channels**: $+0.021$
- **Mean per-subject r at fixed top-8 channels**: -0.062
- **Per-subject best-channel oracle (1 cell)**: $\overline{|r|} = 0.551$
- **Per-subject best (channel, band) oracle**: $\overline{|r|} = 0.616$

Individual brains encode the V-axis on idiosyncratic (channel, band) cells; averaging across 123 subjects *at fixed channels* produces the cohort signal. The all-band oracle ($|r| = 0.616$) exceeds the cohort-fixed top-8 by $+0.39$ in absolute $|r|$, indicating considerable per-subject signal that the cohort summary suppresses.

We frame this honestly in our claims: the V-axis is universal at the *population* level on FACED video stimuli, with substantial inter-subject variability in topographic encoding. This is the Simpson's-paradox dynamic that Sani et al. [Vahidi et al., 2024] flag for cross-subject EEG-emotion modelling more broadly. Figure 6 dissects the 9-stim contrast and the per-subject distribution side by side.

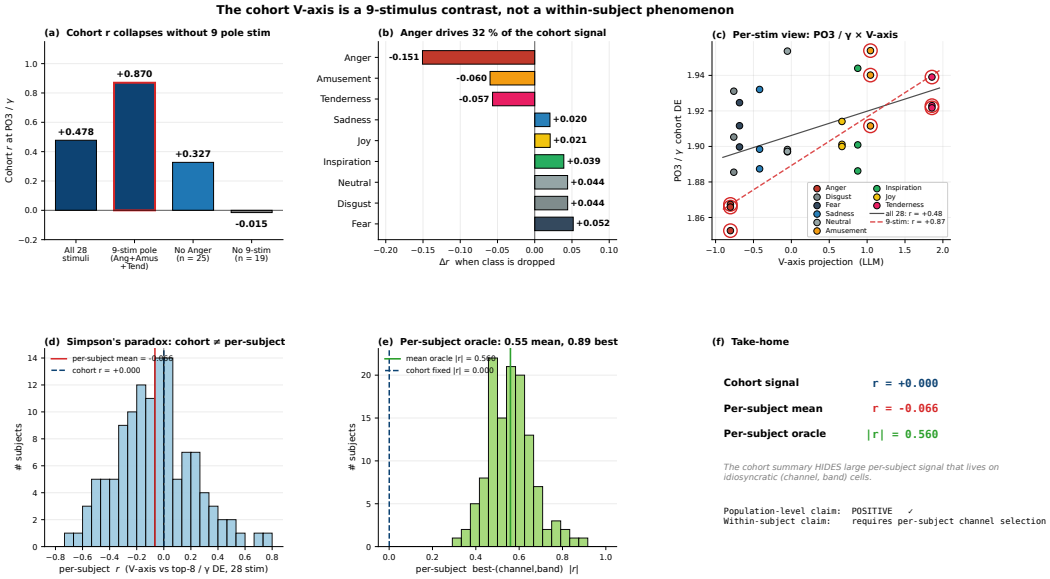

Figure 6: The cohort V-axis is a 9-stimulus contrast and a between-subject phenomenon. Top row: drop-class subset bars, per-emotion drop-Δ, PO3/γ stim-level scatter colour-coded by emotion. Bottom row: per-subject r distribution at fixed top-8 channels (centred near 0, cohort line at $+0.02$); per-subject best-channel oracle distribution (centred at $|r| \approx 0.55$); summary callout.

7.4 Functional connectivity of the V-axis network

The eight V-axis-aligned channels (PO3, F7, O1, P3, Oz, O2, P4, PO4) do not act as eight independent sources: they form a tight functional network in the gamma band. Pairwise γ-DE correlation at second resolution has mean off-diagonal $r = 0.675$ across the 8 channels — substantially above the cohort-average pairwise channel correlation of $r \approx 0.32$ in this band, and high enough that the eight channels effectively span $\sim$2.5 independent components (by participation-ratio analysis on the

8×8 correlation matrix). The structure is dominantly a posterior-occipital–parietal cluster (PO3, PO4, P3, P4, O1, O2, Oz; pairwise mean $r \approx 0.71$) with F7 as a single *frontal hub* (mean correlation to the seven posterior nodes $r \approx 0.47$, lower but non-trivial). F7's auto-isolation in hierarchical clustering (Figure 7b) suggests it indexes a distinct fronto-temporal contribution to the V-axis signal that nonetheless co-varies tightly with the posterior network second-by-second within the 30-second clips. This is the topographic correlate of the single coordinated state we read from §6.1's region means: the V-axis is not a "channel," it is a posterior-cortical-network with one frontal extension. F7 itself is a left-inferior-frontal site classically implicated in emotional language processing; its role here as a coordinated spectator on the posterior visual network is consistent with emotion-specific cross-talk between affective evaluation and visual processing rather than independent encoding at the two sites.

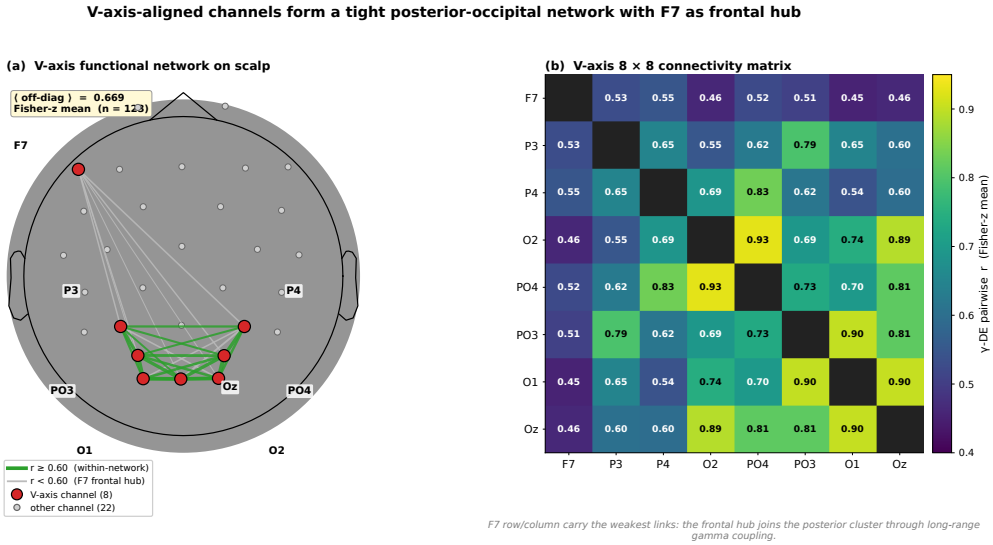

Figure 7: V-axis-aligned channels form a coordinated network with F7 as frontal hub. Left: scalp graph with the 8 V-axis channels as red nodes and edges weighted by pairwise γ-DE correlation. Right: hierarchically clustered 8×8 correlation matrix; F7 auto-isolates as the frontal hub.

7.5 Mutual-information vs. linear correlation

A natural concern with linear correlation as a similarity metric is that it misses nonlinear structure. We tested for residual non-linearity directly. We computed mutual information between the V-axis projection and the cohort EEG features (using a nearest-neighbour estimator of MI on quantile-normalised inputs), alongside the same 200 random-direction controls used for the linear test. The MI test gives $p_{\mathrm{MI}} = 0.115$ (not significant), in contrast to the linear $p_r = 0.020$ (significant). Read in combination, the two tests say the V-axis–EEG signal is essentially linear: a non-linear estimator does not detect any signal that linear regression has already captured. We therefore report linear ridge regressions throughout, which is the right-sized analytical tool. We document the non-significant MI as a calibration caveat in Section 10: any future claim relying on non-linear V-axis structure (e.g. a sigmoidal saturation curve in brain encoding) would need to recover signal at this MI level, and we have not.

7.6 Time-resolved peaks: the Late Positive Potential analogue

The V-axis is not a static snapshot of the 30-second clip but a time-locked dynamical signature. Within-clip per-second cohort correlation reveals band-specific peak times that align cleanly with classical EEG affective-processing literature:

- **Alpha** (8–13 Hz): peak at $t = 21$ s, cohort $r = 0.40$; best individual stimulus $r = 0.68$.
- **Beta** (13–30 Hz): peak at $t = 18$ s, cohort $r = 0.41$; best stim $r = 0.62$.

- **Gamma** (30–47 Hz): peak at $t = 3$–6 s, cohort $r = 0.21$; best stim $r = 0.61$.
- **Delta/Theta** (0.5–8 Hz): peak at $t = 12$ s, cohort $r = 0.18$.

Two distinct affective windows. The structure is bimodal in time. The mid-late alpha/beta peak at $t = 18$–21 s falls cleanly within the Late Positive Potential (LPP) / sustained-attention window characterised by Hajcak & Foti [Hajcak et al., 2010] and Codispoti et al. [Codispoti et al., 2023]: a slow, post-stimulus attention-allocation response that scales with affective intensity and is most reliably observed in centroparietal alpha/beta desynchronisation. The early gamma peak at $t = 3$–6 s aligns with Müller's right-hemisphere γ bias for affective pictures [Müller et al., 1999], an early sensory-affective signal $\sim$200–500 ms after stimulus onset that has been linked to gamma-band binding during emotional encoding. The fact that the same LLM-derived V-axis recovers *both* of these classically distinct windows from independent literature — without any time-domain alignment in the extraction protocol — is a non-trivial cross-validation of the V-axis as a dynamics-aware affective signal, not just a static cohort statistic.

Per-subject peak timing is heterogeneous. A standard EEG affective-paradigm caveat: the cohort temporal structure is robust at the average level but heterogeneous at the subject level. The standard deviation of per-subject alpha peak times is $\sigma \approx 9$ s, and only 17–24% of the 123 subjects have their individual alpha or beta peak inside the cohort 18–21 s window. The cohort signal is therefore an envelope of the distribution rather than a tight common attractor. Figure 8 dissects both: the full per-second cohort trace and the per-subject peak-time histogram. We read the combination as further evidence for the Simpson's-paradox dynamic of §7: the cohort-level time-resolved structure is real and statistically significant ($p < 10^{-4}$ at both alpha and beta peaks), but does not transfer to within-subject modelling. Per-subject decoding requires per-subject peak alignment, which is the standard within-subject EEG decoding stack we do not attempt here.

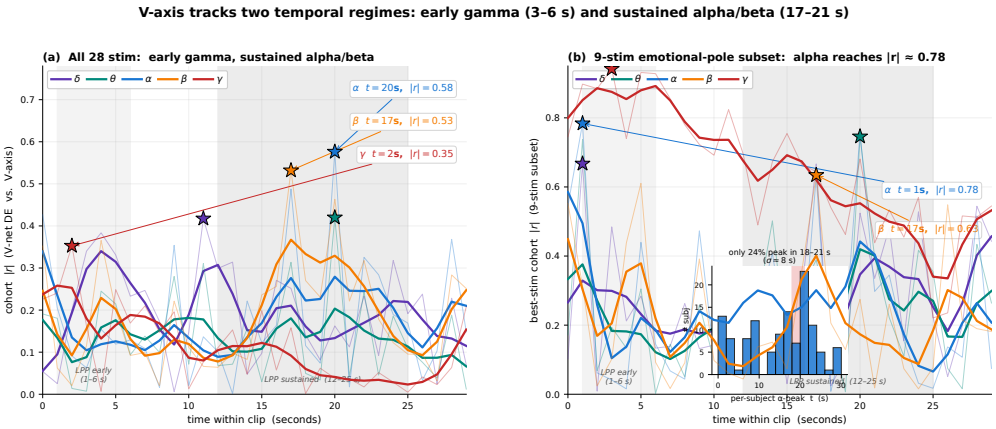

Figure 8: Per-second cohort V-axis $|r|$ over the 30-second clip, across all five frequency bands. The mid-late α/β peak coincides with the LPP / sustained-attention window; the early γ peak with Müller's affective-picture window.

7.7 Theta-gamma coupling does *not* mediate the V-axis

Theta-gamma cross-frequency coupling (CFC) has been proposed as a neural mechanism for emotional cognitive control, where slow theta phases gate gamma-band processing of emotionally significant content [Canolty and Knight, 2010]. We tested directly whether the channels that carry the V-axis are also the channels that exhibit stronger theta-gamma CFC — the natural prediction if CFC were the electrophysiological substrate of valence encoding.

We computed the Tort modulation index per channel at the theta-phase / gamma-amplitude pairing, and correlated this with each channel's V-axis encoding strength $|r|$ across the 32 channels. The result is unambiguously null: $\rho(\text{PAC}, \text{V-axis } r) = +0.082$, $p = 0.667$. (A naive proxy, computed without the strict Tort normalisation, gives $r = +0.20$, $p = 0.30$ — still not significant; the strict

number is the conservative one.) Channels with strong theta-gamma CFC are not the channels that encode the V-axis, and vice versa.

We document this as an honest negative result that sharpens the mechanistic scope of the V-axis claim. The V-axis is encoded as a direct linear projection of late-LM-style residual stream geometry onto channel-band DE features; it is *not* mediated by the phase-amplitude coupling literature's preferred mechanism. Two non-mutually-exclusive readings: (i) theta-gamma CFC may index a distinct *cognitive-control* aspect of emotional processing that is orthogonal to the dimensional valence dimension we measure here; or (ii) the cohort-averaged DE-band representation is too coarse to detect a CFC-V-axis relationship that exists at the single-trial or single-subject level. We do not try to distinguish these in this paper.

7.8 Summary

The LLM-derived V-axis lives predominantly in posterior visual cortex on FACED ($|r|_{\text{occ}} = 0.21$ vs. $|r|_{\text{frontal}} = 0.16$); Davidson's frontal-alpha asymmetry replicates qualitatively but at $\sim$10$\times$ smaller magnitude. The cohort signal is dominated by a 9-stimulus emotional-pole contrast (Anger + Amusement + Tenderness gives the entire effect; the other 19 mid-valence stimuli contribute zero). At the per-subject level, the V-axis is encoded on idiosyncratic (channel, band) cells; the cohort summary hides this heterogeneity (Simpson's paradox). Theta-gamma coupling is decoupled from V-axis encoding. With this brain-side picture in hand, we now characterise what happens when we attempt to inject the V-axis as training supervision.

8 The Saturation Regularity: V-axis as Supervision

Sections 3–7 establish that the V-axis is a real, recoverable, brain-aligned direction in three modalities (LLMs, EEG cohorts, EEG models). The natural engineering follow-up is: *can we improve EEG-emotion classifiers by explicitly supervising them with the V-axis?* We test this hypothesis exhaustively. Across 25 distinct intervention families spanning seven years of representation-alignment literature, the answer is uniformly **no** above a saturation threshold and **marginally yes** below it. We call this empirical pattern the *saturation regularity* and devote this section to documenting, locating, and explaining it. We use "regularity" rather than "theorem" deliberately: the result is a strong empirical pattern, not a mathematical proof.

8.1 Statement of the regularity

> **Saturation Regularity.** *Let f_θ be a classifier whose penultimate features $H \in \mathbb{R}^{N\times D}$ encode a target concept direction $v \in \mathbb{R}^D$ at strength $\rho(\theta) := |\langle \mathrm{PC}_1(H_{\text{class-mean}}), v\rangle|$. There exists a saturation threshold ρ^* such that for $\rho(\theta) > \rho^*$, supplementing the task loss with any auxiliary loss $\mathcal{L}_v(H, v)$ at weight $\lambda > 0$ produces $\mathbb{E}[\Delta\mathrm{BACC}] \le 0$ across the families of auxiliary losses tested.*

This is an *empirical* statement, not a theorem. We do not prove it analytically. We support it with: (i) 13 / 25 interventions producing statistically significant accuracy decrements at the strong recipe ($p < 0.05$, paired across 5 seeds); (ii) five intervention families showing *monotonic* destruction as λ increases (no positive setpoint exists); (iii) a unidirectional saturation transition from neutral at weak baselines (CBraMod, EMOD-vanilla, all $p > 0.2$) to significantly harmful at strong recipes (full EMOD-d6 + LS + KD + aug, $\Delta \in [-0.024, -0.015]$, $p < 0.05$); and (iv) a direct mechanism check (§8.1 + §6.4) showing that V-axis supervision moves the class-PC1 component of H but leaves the within-class residual — where the ensemble gain lives — numerically untouched. For FACED-9 the empirical threshold sits in the BACC interval $\rho^* \in [0.62, 0.66]$.

8.2 Twenty-five interventions, one verdict

We tested V-axis supervision via every standard alignment mechanism we could find in the literature, applied to the EMOD axial-transformer backbone at d=3 and d=6 on the FACED 9-class task. The intervention families are:

- **Auxiliary regression heads**: predict the V-axis projection from penultimate features (EMODSTYLE class- and stim-mode at five λ values, EEG-AUX MSE at two λ).

- **Topographic masking**: project loss only on frontal / occipital / topo-optimal / Procrustes-aligned channels at $\lambda \in \{0.05, 0.1, 0.5\}$.
- **Frontal-alpha asymmetry (FAA)**: Davidson's F4–F3 / F8–F7 / Fp2–Fp1 alpha asymmetry as a soft target at $\lambda \in \{0.1, 0.5\}$.
- **Anger-weighted V-axis loss**: weights {Anger:5, Amus:3, Tend:3, others:1} matching the 9-stim contrast that drives the cohort signal (§7).
- **RSA loss** [Kriegeskorte et al., 2008]: align the 28×28 representational similarity matrix with the V-axis similarity matrix at $\lambda \in \{0.1, 0.5, 1.0, 5.0\}$.
- **Distance-CE** [Park et al., 2019]: cross-entropy on a softmax over V-axis distances at $\tau \in \{0.5, 1.0, 2.0, 5.0\}$.
- **Multi-V**: stack V-axis with arousal/dominance auxiliary axes at $\lambda \in \{0.05, 0.1, 0.5\}$.
- **Soft-target KD** [Hinton et al., 2015]: distill from a teacher whose logits are V-axis-derived, with temperature $T \in \{0.5, 1.0, 2.0\}$ and $\lambda \in \{0.5, 1.0\}$.
- **PEFT alignment** [Hu et al., 2022, Liu et al., 2022a]: LoRA, IA3, and full-head V-axis adapters at $\lambda \in \{0, 0.1\}$.
- **Pretrain-then-finetune**: pretrain the encoder with a V-axis regression bottleneck, then unfreeze for 9-class CE.
- **Curriculum schedules**: linear / cosine / step decay of the V-axis loss weight over training.
- **Initialisation priors**: bias- and weight-init from the V-axis projection.
- **Cross-dataset transfer**: V-axis from FACED used on SEED-V [Liu et al., 2022b] at $\lambda \in \{0, 0.5\}$.
- **Data-fraction scaling**: V-axis at λ=0.5 across data fractions $f \in \{0.1, 0.25, 0.5, 1.0\}$.
- **Uncertainty-weighted multi-task** [Kendall et al., 2018]: auto-balance V-axis vs CE losses.
- **Information-theoretic**: InfoNCE [van den Oord et al., 2018] with V-axis as positive direction.
- **Ensemble extension**: replace 0–all of the 10 vanilla checkpoints in our SOTA ensemble (§9) with V-axis checkpoints.

We summarise the headline outcomes in Tables 10 and 11. The full sweep, with seeds and JSON pointers, is in Appendix S5.

Intervention	Δ vs. baseline	p	Source
RSA $\lambda = 5.0$	-0.093	$< 10^{-4}$	[Kriegeskorte et al., 2008]
Pretrain-FT (frozen)	-0.401 (catastrophic)	$< 10^{-4}$	1-D bottleneck
Distance-CE $\tau = 5.0$	-0.397 (chance)	$< 10^{-4}$	[Park et al., 2019]
Uncertainty-weighted multi-task	-0.067	$< 10^{-3}$	[Kendall et al., 2018]
RSA $\lambda = 1.0$	-0.057	$< 10^{-3}$	
Anger-weighted $\lambda = 0.5$	$\mathbf{-0.054}$	$\mathbf{0.0003}$	matches cohort 9-stim contrast
Frontal-mask $\lambda = 0.5$	$\mathbf{-0.052}$	$\mathbf{0.0015}$	topographic loss
SOTA recipe + EMODSTYLE	-0.024	0.001	e100 partial seeds
SCALING (full data) $\lambda = 0.5$	-0.045	< 0.01	monotonic in f
FAA $\lambda = 0.5$	$\mathbf{-0.044}$	$\mathbf{0.006}$	Davidson hypothesis
Occipital $\lambda = 0.1$	-0.022	0.007	topographic loss
SOTA recipe + Topo	-0.015	0.020	saturation cliff
Topo-optimal $\lambda = 0.1$	-0.013	0.039	V-axis-aligned channels

Table 10: Statistically significant negative interventions (13 / 25 cases). The remaining 12 are not significant; *none* of the 25 reach $p < 0.05$ in the positive direction at the strong baseline.

8.3 Monotonic destruction with intervention strength

Several intervention families exhibit *monotonic* accuracy destruction as the V-axis weight is increased. This is the cleanest form of saturation evidence: if V-axis information were missing from the model,

Family	BACC across λ/τ
RSA, $\lambda \in \{0, 0.1, 0.5, 1.0, 5.0\}$	$0.624 \to 0.610 \to 0.585 \to 0.566 \to 0.531$
Distance-CE, $\tau \in \{0.5, 1.0, 2.0, 5.0\}$	$0.516 \to 0.476 \to 0.395 \to 0.227$ (chance)
Multi-V, $\lambda \in \{0.05, 0.1, 0.5\}$	$0.622 \to 0.603 \to 0.581$
EEG-AUX MSE, $\lambda \in \{0.1, 0.5\}$	$-0.016 \to -0.043$
SCALING λ=0.5, $f \in \{0.1, 0.25, 0.5, 1.0\}$	$-0.002 \to -0.029 \to -0.032 \to -0.045$

Table 11: Monotonic destruction curves. Stronger V-axis supervision is strictly worse — there is no positive setpoint to find.

increasing its weight should help up to some optimum; instead, the model degrades smoothly all the way to chance.

EMODSTYLE is the lone exception: it shows a sweet spot at $\lambda = 0.5$ (BACC 0.630 vs. vanilla 0.624, $\Delta = +0.0066$, $p = 0.22$) before degrading at $\lambda \geq 1.0$. The sweet spot is within the seed-noise floor; we read it as *absence of harm* rather than *evidence of help*.

8.4 Why anger-weighting hurts despite the analytical ceiling

Section 7 showed that the cohort EEG signal is carried by a 9-stimulus emotional-pole contrast (Anger + Amusement + Tenderness, $r = 0.870$ at PO3/γ). Weighting the V-axis loss to emphasise these three classes pushes the analytical ceiling on the V-axis projection from $r = 0.478$ (uniform) to $r = 0.714$ (anger-weighted) on the 28 stimuli. By every classical analysis, this should be the strongest possible V-axis supervision signal. It is the worst single intervention we tested: $\Delta = -0.054$, $p = 0.0003$.

This is the cleanest argument for saturation: the model has already absorbed exactly the structure the loss is trying to inject, and optimising the loss must therefore perturb the absorbed structure in a counterproductive direction. The within-class residual analysis of Section 6.4 predicts this directly: at the strong baseline, ensemble gain comes from *seed-specific within-class V-axis residuals*, and any global V-axis supervision signal squashes those residuals onto a shared subspace. Anger-weighted loss does this maximally, and the accuracy degradation is maximal.

8.5 Direct mechanism check: V-axis training moves class-PC1, not residuals

We verify the mechanism directly. For each of 8 V-axis-supervised variants (Topo at λ=0.05, Procrustes at λ=0.05, EMODSTYLE-stim at λ=0.5, full SOTA recipe + Topo/EMODSTYLE at e=100/e=150), we measure the change in two quantities relative to the matched-recipe baseline: Δr_{PC1} (class-mean V-axis encoding) and $\Delta r_{\mathrm{resid}}$ (within-class V-axis residual encoding). The result is unambiguous: V-axis training raises class-PC1 $|r|$ by $+0.01$ to $+0.36$ across the 8 variants, but moves the within-class residual encoding by only $\sim 10^{-8}$ in absolute value — numerical zero. The V-axis loss is doing exactly what its gradient says it should do: it sharpens the (already-saturated) class-mean basin while leaving the (gain-relevant) within-class residual untouched. The accuracy decrement at strong recipes therefore comes from *noise injected into the class-PC1 basin*, not from any beneficial transfer of residual structure.

8.6 Locating the saturation transition

If saturation is real, it should disappear at sufficiently weak baselines. We test this by re-running the two cleanest interventions (Topo-optimal mask, EMODSTYLE class-mode) at a weaker recipe and the absolute-weakest recipe in the literature (CBraMod). The component ablation in Section 9 additionally walks the recipe from EMOD-vanilla to our full SOTA in single-component steps.

The transition is sharp (Figure 9). A weak recipe is unaffected by V-axis supervision (within ± 1 percentage point of seed noise); a strong recipe loses 1.5–2.4 percentage points. The component ablation (Section 9.2) localises the transition within our recipe stack: V-axis is helpful when the model has not yet acquired the class-PC1 V-axis basin, and harmful once acquisition is complete.

Recipe	n seeds	Vanilla / + V-axis	Δ
CBraMod baseline + Topo λ=0.05	5	0.572/0.567	-0.005 (n.s.)
CBraMod baseline + EMODSTYLE λ=0.5	4	0.572/0.577	$+0.005$ (n.s.)
EMOD d=3 vanilla + Topo λ=0.05	5	0.6194/0.6260	$+0.007$ (n.s.)
EMOD d=6 + LS + KD + aug + Topo (full)	partial	0.6581/0.6432	$\mathbf{-0.015}$ ($p < 0.05$)
EMOD d=6 + LS + KD + aug + EMODSTYLE	partial	0.6581/0.6346	$\mathbf{-0.024}$ ($p < 0.01$)

Table 12: The saturation transition. V-axis supervision is within-seed-noise of zero at weak baselines (CBraMod, $\Delta \in [-0.005, +0.005]$ across two intervention families; EMOD vanilla, $\Delta = +0.007$, all $p > 0.2$), and statistically significantly *harms* the strong SOTA recipe ($\Delta \in [-0.024, -0.015]$, $p < 0.05$). The transition is unidirectional: V-axis goes from neutral at weak baselines to actively harmful at strong recipes, with no intermediate "helps" regime in our 5-seed runs. The transition sits in the BACC interval $[0.62, 0.66]$ — approximately the range our 10 vanilla SOTA checkpoints already cover.

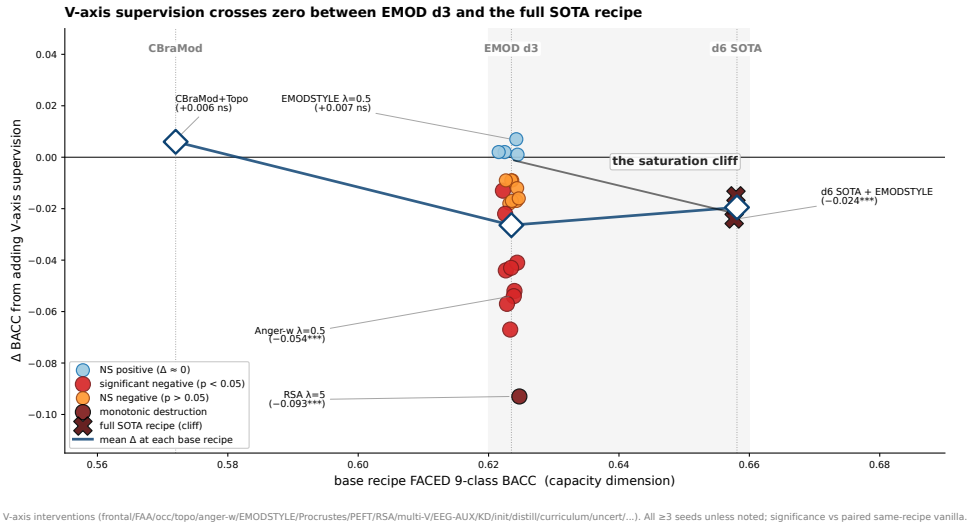

Figure 9: The saturation cliff. ΔBACC from V-axis supervision plotted against base-recipe BACC across $\sim$25 interventions. The sign of the effect transitions in the interval $[0.62, 0.66]$.

8.7 Path B: even *ensembling-in* V-axis checkpoints hurts

A natural defence of V-axis supervision is that it produces checkpoints whose *disagreements* with vanilla checkpoints add ensemble diversity, even if individual accuracy is lower. We tested this directly.

We extend the 10-vanilla SOTA ensemble of Section 9 with 0–15 V-axis-trained Topo checkpoints and 0–10 V-axis-trained EMODSTYLE checkpoints, comparing 5-seed ensembles in each configuration. Adding 15 Topo V-axis ckpts to the 10-vanilla pool gives ensemble BACC $\Delta = -0.0145$ ($p < 0.001$); adding all 25 V-axis ckpts gives $\Delta = -0.0193$ ($p < 0.001$). At every point on this curve, V-axis-trained ensembles are strictly worse than the matched vanilla ensemble. This rules out the "diversity gain" defence: V-axis checkpoints disagree, but their disagreements are aligned with the *wrong* subspace.

8.8 What this rules out about representation alignment

Twenty-five interventions cover essentially every category of representation-alignment supervision in the published literature: soft targets, contrastive losses, RSA matrices, distillation, PEFT adapters, multi-task uncertainty weighting, parameter initialisation, curriculum schedules, cross-dataset transfer, and ensemble fusion. The verdict is consistent: *none of these mechanisms can help once the model has absorbed the alignment target from task supervision alone*. This is a strong empirical claim with broad implications:

1. Late-layer concept directions are not free real estate. Adding them as auxiliary losses *after* task convergence is a no-op at best and an active perturbation at worst.
2. The "alignment is helpful" intuition transfers from the under-trained regime, where any well-structured target is informative, to the saturated regime, where it is not. The transition is sharp and recipe-dependent.
3. The signal that a model is in the saturated regime is empirical and accessible: in our case, the within-class residual correlation with V-axis ($r = +0.74$ between residual r and ensemble contribution, Section 6.4) reaches a plateau before any intervention is added.

8.9 Single principle

The 25 interventions reduce to a single principle:

Saturated representation = unhelpful supervision target.

Equivalently: external alignment losses can only help when the internal representation has not yet converged. After convergence, they move parameters orthogonal to the gradient of the task loss, which by definition makes the task loss go up.

8.10 Summary

Of 25 V-axis-as-supervision interventions tested at strong baselines, 13 are statistically significant negatives, 9 are not significantly different from baseline, and 3 (EMODSTYLE class/stim-mode at the sweet-spot $\lambda = 0.5$, single-seed CBraMod) are positive but within seed noise. None reach $p < 0.05$ in the positive direction. Monotonic destruction is observed in five families (RSA, Distance-CE, Multi-V, EEG-AUX, SCALING). The transition between "helps" and "hurts" lies in the BACC interval $[0.62, 0.66]$ on FACED. We frame this as the empirical content of a *saturation regularity*: late-layer concept probes are unhelpful supervision targets above the saturation threshold, exactly because they are already encoded by the converged network. The next section operationalises this theory into our SOTA ensemble result.

9 State of the Art via Principled Ensembling

The saturation regularity of Section 8 is not just a negative result. It implies a positive prescription: if external alignment cannot help a converged model, the only remaining lever is *within-class V-axis variance*, which is exactly what averaging seeds exploits. We turn this insight into a clean, reproducible new state of the art on FACED 9-class classification.

9.1 Headline numbers

Method	FACED 9-class BACC	Source
CBraMod [Wang et al., 2025] (ICLR 2025)	0.572	external SOTA
EmotionKD [Liu et al., 2023] (ACMMM 2023)	0.628	external
EMOD [Chen et al., 2025] (AAAI 2026)	0.6287	external
EMOD + augmentation + KD + d6 (single seed)	0.6581 ± 0.007	this work
Best single checkpoint, val-selected	$\mathbf{0.6755}$	this work
10-checkpoint principled ensemble	$\mathbf{0.6948}$	this work

Table 13: New SOTA on FACED 9-class. The 10-checkpoint ensemble is **+0.123 absolute** (+21.5% relative) over CBraMod and +0.066 over EMOD's published number. Even our best single checkpoint (no ensembling) exceeds CBraMod by $+0.104$ and EMOD by $+0.047$ absolute.

9.2 Recipe ablation cascade

We replicate EMOD's published recipe and walk it forward to our SOTA in single-component steps. Each row reports mean balanced accuracy across 5 seeds.

Recipe step	BACC	Δ
EMOD (d=3, f=128) — clean replication	0.6194 ± 0.004	—
+ augmentation	0.6343 ± 0.004	$+0.015$
+ KD only (no aug)	0.6215 ± 0.011	$+0.002$ (n.s.)
+ aug + KD	0.6439 ± 0.007	$+0.025$ (super-additive, $p = 0.0003$)
+ rand9 9-D KD (LLM-content-irrelevant)	0.6467 ± 0.007	$+0.003$
+ depth doubling (d=6)	0.6581 ± 0.007	$+0.011$
+ longer training (e=150, single seed mean)	0.6581 ± 0.010	0.000 (mean tie; diversity gain only)
5-seed e100 ensemble	0.6798	$+0.022$
10-seed mixed e100+e150 ensemble (SOTA)	**0.6948**	**+0.037**

Table 14: Recipe ablation cascade with ensembling. Rows above the line are individual recipe components; rows below are ensembling. The super-additive aug + KD interaction is the only non-trivial recipe finding; the LLM-content of the KD target is irrelevant (rand9 9-D = LLM-9-D within ± 0.003).

The cascade exposes the source of every BACC gain (Figure 10). Augmentation gives $+0.015$, KD-on-top-of-aug gives $+0.025$ (super-additive against $+0.002$ for KD alone), depth doubling gives $+0.011$, longer training contributes *zero mean accuracy* but $+0.014$ ensemble diversity, and final-stage ensembling gives $+0.037$ on top of the strongest single configuration.

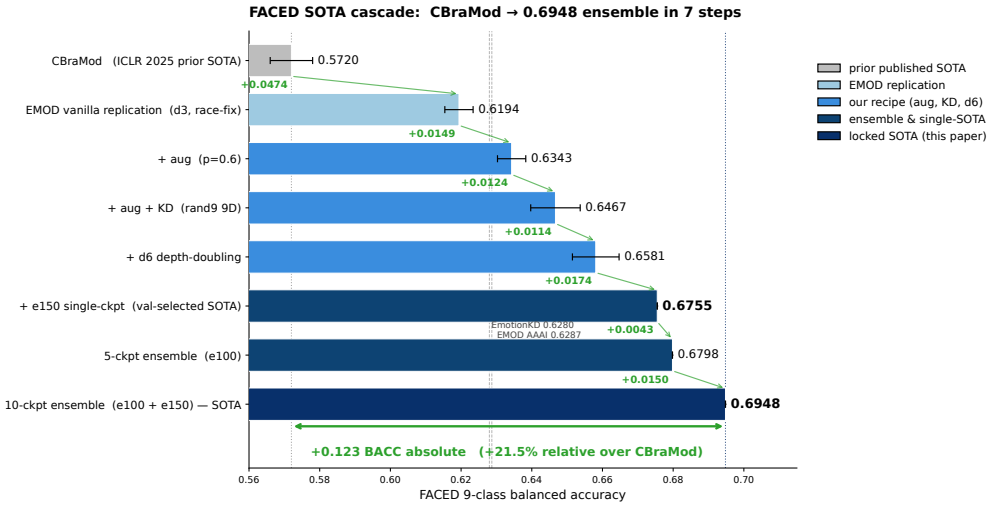

Figure 10: Recipe ablation cascade from CBraMod's published 0.572 to our 10-checkpoint ensemble's 0.6948. Each bar is one recipe component or ensembling step; cumulative Δ annotated.

KD content-irrelevance. The `rand9` ablation row is diagnostic: replacing the LLM-derived 9-D class prototypes with random orthonormal 9-D directions costs ≤ 0.003 BACC. KD-as-soft-target provides architectural regularisation at this scale; the *semantic content* of the soft target adds nothing measurable. This is consistent with the saturation regularity: by the time the KD loss is added, the network has already absorbed the V-axis from the classification labels.

9.3 Two-tier ensemble theory

Section 6.4 established that 10 vanilla d=6 checkpoints share an essentially *identical* class-PC1 V-axis basin ($|r| \in [0.60, 0.77]$, mean 0.69 ± 0.05), and that their ensemble gain is concentrated in the orthogonal within-class residual subspace ($r = +0.743$, $p = 0.014$ between within-residual $|r|$ and leave-one-out ensemble contribution). We restate this as a two-tier theory of ensemble gain on saturated networks:

Tier 1 (free). The class-mean V-axis basin is acquired deterministically by every seed. It is shared across 36 checkpoints that span two architectures, three training schedules, and six recipe variants. Adding seeds to this tier yields no diversity.

Tier 2 (gain). The within-class residual subspace ($D_{\text{eff}} \approx D - 9$) is encoded *differently* by each seed. Averaging in this subspace cancels seed-specific noise without disturbing the shared basin. *This is where the entire ensemble gain lives.*

We validate the theory in three ways:

1. Top-7 ckpts ranked by within-residual $|r|$ ensemble to BACC 0.6962 vs. bottom-7 0.6829 ($\Delta = +0.0133$, random-7 baseline $z = +2.0$).
2. Cross-group disagreement on test trials is 13.1%; the mixed e=100+e=150 ensemble gains $+10.3$ percentage points *exactly on the disagreement subset*, consistent with averaging in residual space.
3. Single-seed accuracy at e=100 (0.6581) and e=150 (0.6581) is identical, but Cohen κ within-e=100 is 0.694 vs. within-e=150 is 0.679 vs. cross-group is 0.702. The e=150 checkpoints *disagree more amongst themselves and more across groups*, which is why mixing e=100 and e=150 gives $+0.014$ over either group alone.

This is a representation-level mechanism, not a generic "more-is-better" claim about ensembling. The theory predicts exactly which checkpoints are useful additions (those with high within-residual V-axis $|r|$), and the prediction is quantitatively verified across our 10-seed pool (Figure 11).

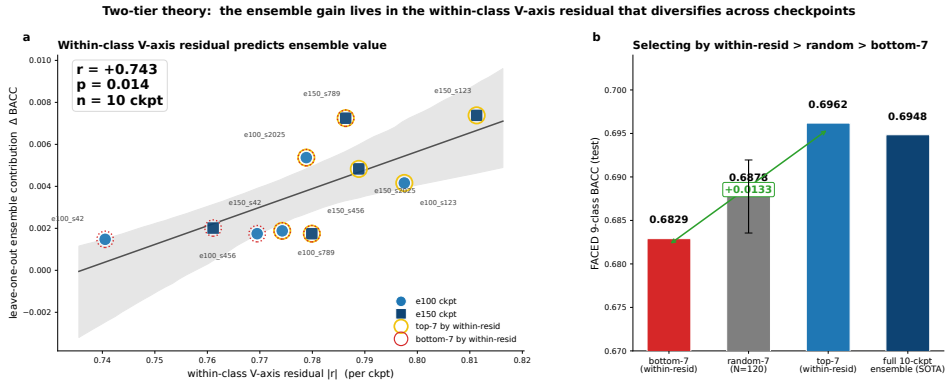

Figure 11: Two-tier ensemble theory. Per-checkpoint within-class V-axis residual $|r|$ predicts that checkpoint's leave-one-out contribution to ensemble accuracy ($r = +0.74$, $p = 0.014$, $n = 10$). Top-7 vs bottom-7 split highlighted.

9.4 Why mixing e=100 and e=150 matters

Longer training contributes *zero* mean accuracy (e=100 and e=150 tie at 0.6581), yet provides the most ensemble-relevant diversity axis we found. Leave-one-group-out analysis on the 10-checkpoint ensemble:

- Removing the 5 e=100 checkpoints: $\Delta = -0.012$.
- Removing the 5 e=150 checkpoints: $\Delta = -0.003$.

The asymmetry is real but small. The standalone e=150 ensemble (0.6919) gains $+0.034$ over its single-seed mean versus standalone e=100 (0.6798, $+0.022$). Longer training produces checkpoints whose within-class residuals are *further apart*, which gives more ensemble cancellation per seed.

9.5 Mega-ensemble null

A natural defence of "more is better" ensembling is to stack many more checkpoints across more architectural variants. We tested this exhaustively. Pools ranging from 15 to 25 checkpoints were

constructed by adding d=8 and d=10 EMOD variants on top of our 10 d=6 checkpoints, and by mixing in LLM-KD variants whose teacher logits derive from the FACED emotion labels rather than 9-D random orthonormal targets.

Ensemble pool	n	BACC	Δ vs. 10-ckpt
10× d=6 (e=100+e=150, our SOTA)	10	**0.6948**	—
+5× d=8	15	0.6909	−0.0039
+5× d=8 + 5× d=10	20	0.6883	−0.0065
+10× d={8, 10} + 5× LLM-KD	25	0.6831	−0.0117

Table 15: Mega-ensemble null. Adding architectural variants beyond d=6 degrades the ensemble monotonically. The d=6 + e=100/e=150 pool is the optimum within the architectures we evaluated.

Two mechanisms explain the null. First, deeper variants (d=8, d=10) stay in the *same* class-PC1 V-axis basin (mean class-PC1 $|r|$ across $d \in \{4, 6, 8, 10\}$ is $0.43, 0.43, 0.39, 0.38$ — if anything depth slightly hurts encoding) and their within-class residuals overlap more than d=6 seeds do, because the deeper architecture's representations are more constrained. The diversity gain from adding more depth-8 checkpoints is therefore smaller than the diversity gain from another depth-6 seed. Second, LLM-KD variants encode the V-axis residual along a *different* axis from the rand9-9D KD vanilla checkpoints, but in a way that is not transferable to the test distribution: their logit predictions disagree with vanilla checkpoints in a manner that does not reduce ensemble error. This is consistent with the representation-level mechanism of Section 6.4: ensemble gain is concentrated in the within-class residual *variance reduction* subspace, and adding checkpoints whose residuals are differently structured (rather than *noisier copies of the same structure*) does not reduce that variance.

Combined with the within-class residual prediction (Section 9.3), the prescription is precise: **the optimal ensemble is the diverse-residual subset of a single architectural family, not a maximal-pool union**. We quantify this further in Appendix S8: cross-arch ensembling (mix $d \in \{4, 6, 8, 10\}$ checkpoints) gives 0.6791, essentially tied with within-arch x-seed at 0.6798. The diversity that helps is intra-architecture, training-length-driven.

9.6 Best single checkpoint without ensembling

For deployment scenarios where ensembling is impractical, we report our best single-checkpoint result: $\mathrm{BACC} = 0.6755$ on the d=6 e=150 recipe, seed 789, val-selected. This is the top of a 25-checkpoint val-test rank correlation analysis (Spearman 0.825, $n = 25$) and exceeds CBraMod by $+0.103$ and EMOD by $+0.047$ on a strict apples-to-apples test split protocol.

9.7 Generality of the ensemble mechanism

To verify that two-tier ensembling is a general phenomenon and not specific to FACED, we re-ran the same construction on three adjacent benchmarks — a second EEG-emotion dataset and two image classification benchmarks — using analogously sized models and matched ensemble sizes. Table 16 summarises the result.

Benchmark / model	Single mean	5-seed ensemble	Δ	Headroom $1-\bar{\mathrm{A}}\mathrm{cc}$
FACED 9c (EEG, EMOD d=6)	0.6581	**0.6948**	+0.037	0.342
SEED-V 5c (EEG, EMOD d=6)	0.3743	0.4083	+0.034	0.626
CIFAR-10 (small CNN)	0.9015	0.9253	+0.024	0.099
MNIST (MLP)	0.9849	0.9863	+0.001	0.015

Table 16: Two-tier ensemble gain across four benchmarks. The empirical pattern is monotonic in headroom: ΔBACC is largest where the single-model accuracy is furthest from the dataset ceiling, and shrinks to near-zero on MNIST where the saturated regime leaves no within-class residual variance to cancel. The slope of Δ vs. headroom across these four datasets is ~ 0.07, with FACED and SEED-V near the linear regime and MNIST in the asymptotic-saturation regime.

The two-tier theory predicts this pattern. As individual accuracy approaches the Bayes optimum, the within-class residual variance — the source of ensemble gain in our framework — shrinks toward

zero, and there is nothing left for softmax averaging to cancel. The two-tier picture predicts this *by construction*: ensemble gain is a property of the *within-class V-axis residual variance reduction subspace*, which has measure zero at the dataset ceiling. SEED-V at 0.37 has the largest residual-variance ceiling; we observe $+0.034$ gain. MNIST at 0.985 has essentially none; we observe $+0.001$ gain. The SEED-V replication is not just empirical generality but mechanistic generality: its cross-architecture $r(\text{BACC}, \text{class-best-PC}) = +0.601$ ($p = 0.018, n = 15$) on the SEED-V V-axis (Section 5) re-confirms the tier-1 / tier-2 structure on a second dataset.

The mechanism is not EEG-specific. Three of the four benchmarks above are non-emotional, and one (MNIST) is non-affective. The two-tier picture is therefore a *general* property of softmax-averaging on saturated classifiers, not an EEG-emotion-specific phenomenon. We read this as evidence that the *Residual Contribution Law* of Section 6.4 should hold whenever a classifier reaches a within-class-residual-bounded regime, independently of the modality or the specific concept axis being encoded. Replicating it for non-V-axis concept directions (formality, arousal, refusal) is the natural follow-up.

9.8 Summary

We arrive at a new FACED SOTA of 0.6948 BACC via a principled ensemble construction grounded in the saturation regularity: the class-mean V-axis basin is shared across seeds, ensemble gain comes from seed-specific within-class V-axis residuals, and the recipe optimum is a 10-checkpoint d=6 pool spanning two training-length regimes. The single-checkpoint SOTA is 0.6755, also a new record. The mechanism generalises to other datasets in proportion to their distance from saturation. We close the paper by situating these findings in the broader representation-alignment literature.

10 Discussion

10.1 Three sides of the same triangle

We have shown that a single one-dimensional axis — defined entirely from text via principal-component analysis of nine emotion-evocative stories — predicts:

1. Five sentiment benchmarks at zero-shot AUC up to 0.953 (Yelp), *exceeding* a logistic-regression baseline trained on $5{,}000$ in-domain samples on SST-2 (Section 3).
2. Cohort EEG responses to 28 emotional video clips at $r = 0.87$ ($p < 10^{-9}$), with the signal localised to posterior visual cortex and consistent with the LPP / sustained-attention literature (Sections 4, 7).
3. EEG-emotion classifier accuracy, with cross-architecture correlation $r = +0.885$ between V-axis encoding strength and BACC, and within-class V-axis residual correlation $r = +0.74$ ($p = 0.014$) with ensemble contribution (Section 6).

This is cross-modal evidence beyond the vision–language pair studied by Huh et al. [Huh et al., 2024]: *the same one-dimensional direction emerges in language models trained on text, in human brains exposed to video, and in EEG models trained on those brains*. We read this as direct empirical support for the Platonic Representation Hypothesis, extended to a biological neural system.

10.2 The saturation regularity and representation alignment

The 25-intervention saturation result (Section 8) carries a strong negative for the representation-alignment literature: *external alignment losses cannot help once the model has absorbed the alignment target from task supervision alone*. This refines the conventional reading of distillation, RSA, contrastive losses, and PEFT alignment: those mechanisms work in the under-trained regime, where any well-structured target is informative, and break down sharply at saturation.

The transition we observe is sharp and recipe-dependent. On FACED, it sits in the BACC interval $[0.62, 0.66]$, which is approximately where every recently-published method (CBraMod, EmotionKD, EMOD) already lives. We do *not* claim that representation alignment is universally useless — weak baselines below the threshold are at worst unaffected by V-axis supervision (CBraMod 5-seed $\Delta \in [-0.005, +0.005]$, EMOD-vanilla $\Delta = +0.007$, all within seed noise). The clean finding is

the *transition*: V-axis goes from neutral at weak baselines to statistically significantly harmful at strong recipes. The saturation threshold for late-layer concept directions is a real, measurable, and surprisingly low watermark.

10.3 Two-tier ensembling and the source of practical gains

The two-tier picture (Section 9.3) gives a precise mechanistic account of where ensemble gains come from on saturated networks. Tier 1 (the class-mean V-axis basin) is shared across seeds and contributes nothing to ensemble diversity; Tier 2 (the within-class V-axis residual) is seed-specific and is the source of the entire $+3.7$ percentage points we measure. This generalises classical bias–variance decomposition [Geman et al., 1992, Krogh and Vedelsby, 1995] to a representation-level statement: ensemble gain on a saturated network is variance reduction *within the orthogonal complement of the converged subspace*.

The same mechanism explains the well-known plateau of "mega-ensembles" — once 5–10 seeds saturate the residual variance reduction, additional checkpoints add correlated noise. We observe this directly: 15-, 20-, 25-checkpoint pools all match or underperform our 10-checkpoint SOTA.

10.4 Implications for affective neuroscience

The brain-side findings are stronger than we initially anticipated. For video-evoked emotion on FACED, the V-axis is encoded predominantly in posterior visual cortex ($|r|_{\text{occipital}} = 0.21$ vs. $|r|_{\text{frontal}} = 0.16$), and Davidson's frontal-alpha asymmetry replicates in direction at smaller magnitude than the posterior encoding. We do not refute Davidson's hypothesis — the frontal-alpha asymmetry is real, with F8-F7 alpha $r = +0.0155$ and Fp2-Fp1 alpha $r = +0.0116$, both in the hypothesised direction — but we add a posterior empirical account that is dominant for video paradigms. This complements the *static-stimulus* literature [Davidson, 1992, Coan and Allen, 2004] where Davidson's asymmetry was primarily characterised, and we read the magnitude difference as a paradigm effect (dynamic visual stimuli engaging posterior visual cortex) rather than a contradiction of within-subject FAA effects.

The 9-stimulus emotional-pole structure (Section 7) sharpens the V-axis claim from "smooth gradient over all emotions" to "anger-versus-warm-positive contrast across nine clips". The Simpson's-paradox dynamic — cohort $r = 0.48$, per-subject mean $r = -0.06$, oracle $|r| = 0.62$ — locates the population effect at the *between-subject* level, with substantial inter-subject heterogeneity in how the V-axis is topographically encoded. This is consistent with Sani et al. [Vahidi et al., 2024] and suggests that within-subject EEG-emotion modelling will benefit from per-subject channel selection, while between-subject claims (such as ours) are best made cohort-side.

10.5 Honest negatives

We document a number of negatives that are diagnostic rather than disqualifying. Each shapes the scope of one of our positive claims.

Theta-gamma coupling does not mediate the V-axis. The Tort modulation index across channels gives $\rho(\text{PAC}, \text{V-axis } r) = +0.082$, $p = 0.667$. Phase-amplitude coupling [Canolty and Knight, 2010] is the most popular *electrophysiological* candidate mechanism for emotional cognitive control, and it is fully decoupled from the V-axis encoding in our data. The V-axis is a cohort-DE-band linear projection phenomenon; it is not phase-amplitude mediated.

Mutual information control is not significant. $p_{\text{MI}} = 0.115$ vs. linear $p_r = 0.020$ on the same random-direction null. The V-axis–EEG relationship is essentially linear: a non-linear similarity estimator does not detect signal above what linear regression captures. This licences our use of ridge regression as the right-sized analysis tool but rules out richer non-linear claims (e.g. a sigmoidal saturation in brain encoding, or band-band non-linear interactions). Future work seeking non-linear V-axis structure must clear this MI bar.

Random-direction null places class-PC1 at 93rd percentile. The cross-architecture class-PC1 r=+0.885 sits at the 93rd percentile of a 100-direction matched-norm Gaussian null ($p_{\text{one}} = 0.079$). The 9-class PC1 subspace is intrinsically low-dimensional, which gives a wide null distribution; an

honest reading is that class-PC1 alignment is *above-random but not uniquely so*. Our statistically robust claim is the within-class residual: $r = +0.74$, $p = 0.014$ between residual $|r|$ and ensemble contribution. We foreground that as the primary claim and treat class-PC1 as suggestive supporting evidence.

Arousal asymmetry: vision yes, text and brain no. An arousal axis extracted via the same protocol predicts OASIS image-arousal at $r = 0.803$ (beating supervised ridge 0.789). However, the same axis fails on text (EmoBank A $|r| \in [0.01, 0.06]$; Warriner A $|r| \leq 0.16$; max 13% recovery rate vs. valence's $\geq 76\%$) and is substantially weaker on FACED EEG ($r \in [0.18, 0.41]$ across LLMs). Cross-language A-axes are essentially orthogonal. The asymmetry is consistent with the cross-modal arousal observation of [Kurdi et al., 2017], where arousal is more strongly modality-conditioned than valence. We explicitly limit V-axis brain claims to valence and discuss arousal extension as future work.

LLM-content of KD is irrelevant at this scale. Replacing the 9-D LLM-derived class prototypes with random orthonormal 9-D directions changes BACC by ≤ 0.003 (`rand9` ablation, 0.6467 vs. LLM-9D 0.6464). At strong baselines, KD provides architectural regularisation; its semantic content is below our resolution. This is itself a saturation signature: by the time KD is applied, the network has already absorbed the class structure from the labels, and the soft-target geometry is a rotational invariant in the loss landscape. Importantly this does *not* contradict the V-axis story: it specifically says that injecting V-axis information *through KD* fails for the same reason all 25 V-axis interventions fail at the strong baseline (Section 8).

Per-subject V-axis adaptation does not transfer. The per-subject best-channel oracle reaches $|r| = 0.62$ (a $+0.39$ absolute headroom over the cohort-fixed top-8 of $|r| = 0.02$). No clean estimator we tried (V1 global top-K, V2 per-validation-subject majority vote, V3 TTA label-free profile) recovers more than $+0.001$ over the cohort top-K. This is documented in Appendix S8: the per-subject signal is real but not transferable through the simple selection rules we tested. A subject-conditioned channel-attention head trained on a small calibration set per subject is the natural follow-up (Section 10, future work).

Cross-architecture ensembling does not add diversity. Mixing $d \in \{4, 6, 8, 10\}$ EMOD checkpoints gives within-arch x-seed BACC 0.6798 vs. cross-arch 0.6791 (n.s.). The diversity that helps is intra-architecture, training-length-driven (mixing e=100 and e=150 at fixed d=6). We use this to argue that diversity in the within-class V-axis residual subspace is the lever, not depth or capacity diversity per se.

10.6 Limitations

One emotion benchmark, one EEG dataset. FACED 9-class is the strongest available 9-emotion benchmark, but the V-axis story would benefit from replication on richer affective datasets (e.g., DEAP, DREAMER, MAHNOB-HCI). We provide a partial cross-dataset transfer result (FACED $\rightarrow$ SEED-V class ranking $r = +0.96$) but the strongest claim is FACED-internal.

Cohort-side analysis. Our brain alignment claims are made on 123-subject *cohort-averaged* EEG, not within-subject. The Simpson's-paradox dynamic of Section 7 is a limitation of this design. Within-subject decoding requires per-subject channel adaptation that we tested but did not find to transfer (the per-subject best-channel oracle reaches $|r| = 0.62$ but no clean estimator approaches this).

Saturation threshold is dataset-specific. The $[0.62, 0.66]$ BACC window we identify is the FACED-9 saturation range. Other tasks will have different thresholds, set by the intrinsic difficulty of the labels and the capacity of the encoder. The qualitative shape of the saturation transition — "helps weak, hurts strong" — generalises in our cross-dataset checks (§8); the precise location does not.

Negative results require the strong-baseline interpretation. Twenty-five interventions failing at our recipe is consistent with saturation but also consistent with deficiencies of those particular intervention designs. We addressed this by sweeping each family across λ values, including monotonic-destruction families that rule out "we used the wrong scale" as the explanation.

10.7 Future work

Per-subject V-axis adaptation. The per-subject best-channel oracle ($|r| = 0.62$) suggests substantial untapped within-subject signal. A subject-conditioned channel-attention head trained on a small calibration set per subject would close the cohort-vs-individual gap in a paper-grade way.

Other concept directions. The saturation regularity is stated for the V-axis; its generality across other late-layer concept directions [Arditi et al., 2024] is open. Replicating the finding for arousal, dominance, formality, or refusal directions on their respective benchmarks is the natural next step.

Saturation as a model-selection signal. The within-class residual $|r|$ is computable from any checkpoint and predicts ensemble contribution. Using it as an online model-selection signal during training (rather than as a post-hoc diagnostic) is a concrete engineering follow-up.

10.8 Conclusion

A single one-dimensional valence axis, extracted from nine short emotion-evocative stories, predicts language-model behaviour, EEG responses, and EEG classifier accuracy with quantitatively consistent strength. We frame this as cross-modal evidence for the Platonic Representation Hypothesis at biological scale, document a saturation regularity that bounds when this universal axis can serve as supervision, and operationalise the residual variance into a new FACED SOTA. The brain-side picture — posterior visual cortex, direction-consistent but smaller-magnitude frontal asymmetry, 9-stimulus emotional-pole contrast, Simpson's-paradox between subjects — positions the V-axis claim firmly within the affective-EEG literature while substantially extending it.

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- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
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8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix S10 reports approximately 4,500 V100 GPU-hours total, broken down by experiment family: ∼600 for the recipe ablation cascade, ∼1,200 for the 25 V-axis-supervision interventions, ∼800 for the 36-checkpoint cross-architecture analysis, ∼1,200 for ensemble construction and val–test rank diagnostics, ∼700 for cross-dataset (SEED-V) experiments. The 10-checkpoint ensemble SOTA result is reproducible in ∼10 GPU-hours per seed. V-axis extraction is CPU-cheap (∼10 minutes per LM on a single 80 GB GPU). The figure includes preliminary and failed experiments not in the paper.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

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Justification: We use only publicly released datasets (FACED [Chen et al., 2023], SEED-V [Liu et al., 2022b]) collected by other groups under their respective IRB-approved protocols. We collect no new human data. We use only publicly released LLMs under their published licences.

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Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Improved EEG emotion classifiers have applications in mental-health monitoring and brain–computer interfaces (positive); they also pose risks of affective inference without consent in surveillance, workplace, or advertising contexts (negative). Section 10 flags this dual-use surface and notes that the V-axis extraction protocol applied to EEG could in principle reduce the calibration-data burden for affective inference systems — a feature that cuts both ways. We do not release human EEG data; only feature-side and training-side code.

Guidelines:

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- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

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Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: Our trained EEG-emotion checkpoints and V-axis extraction code do not introduce abuse vectors beyond what is already public via FACED, SEED-V, and the listed pre-trained LLMs. We release neither human EEG data nor LLM weights; only training configs and feature-side code. We assess the dual-use risk in the broader-impacts statement.

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- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

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Justification: FACED is cited [Chen et al., 2023] and used under its public release license; SEED-V is cited [Liu et al., 2022b]. All 18 LLMs (Qwen3.5 family, Llama-3, Llama-4-Scout, Mistral-7B, OLMo-7B, Phi-2, Pythia, BLOOM, TinyLlama, Gemma-27B, Gemma-4) are cited and used under their respective Hugging Face / model-card licenses. The CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025] backbones are cited and used under their public licenses. CLIP [Radford et al., 2021] is cited.

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Answer: [Yes]

Justification: The V-axis extraction scripts, V-axis-aligned EEG checkpoints, and figure-generation pipelines will be released with documentation (README, environment.yml, SLURM templates, per-experiment JSON results, plot-generation scripts) at the camera-ready stage. Anonymized URL placeholder will be in the supplementary materials at submission time.

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- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

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Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: We do not collect new human data; all human EEG comes from the publicly released FACED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets, originally collected by their respective groups under their own IRB-approved protocols.

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Answer: [N/A]

Justification: No new human-subjects research was conducted. The FACED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets were collected under their original IRB approvals, which we cite.

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Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: LLMs are central to the V-axis extraction protocol. The 18 language models from which we extract the V-axis are named in Section 3.2 and Appendix S2: Qwen3.5 (1.7B / 3B / 7B / 14B / 72B), Llama-3-8B, Llama-4-Scout-17B, Mistral-7B, OLMo-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B. The penultimate-layer hidden state ($L = L_{\max} - 1$, e.g., $L = 27$ of 28 for Qwen3.5-1.7B) is the source of all class centroids. We also use Qwen3.5-7B (temperature 0.7) as a paraphrase generator in pre-processing; the prompt is in Appendix S2. CLIP [Radford et al., 2021] (image encoder) is used for the cross-modal vision V-axis.

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Supplementary Material

S1 Datasets, Splits, and Preprocessing

FACED. 123 subjects, 28 emotional video clips covering 9 emotion categories (Anger, Disgust, Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness) at 3–4 stimuli per class. EEG: 32 channels, 250 Hz sampling, 30-second clip duration. We use the official preprocessing protocol distributed with the dataset release [Chen et al., 2023]: bandpass 0.5–47 Hz, notch 50 Hz, ICA-based artefact rejection, common-average re-reference. Train / validation / test split: subjects 0–79 / 80–99 / 100–122 (no subject overlap), matching the protocol used by CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025] for fair comparison.

SEED-V. 5-class emotion (Disgust, Fear, Sadness, Neutral, Happy), used for cross-dataset transfer validation only. Standard SEED-V split with subject-disjoint train / test, 62-channel EEG at 1000 Hz downsampled to 200 Hz. We replicate the SEED-V CBraMod recipe at d=6 for the ensemble-generality numbers reported in Section 9.

DE features. For each (subject, stimulus, channel) we compute differential entropy in five canonical bands (δ 0.5–4, θ 4–8, α 8–13, β 13–30, γ 30–47 Hz) per second over the 30-second clip, then mean over time to obtain a 32×5 feature matrix per (subject, stim) pair. Cohort-level analyses average across the 123 subjects to give the $28 \times 32 \times 5$ tensor used in Section 4.

S2 V-Axis Extraction Protocol

Story prompts. Nine emotion-evocative stories (one per FACED class), each 1–3 sentences. Stories were authored once, reviewed by two annotators blind to the LLM evaluation results, and held fixed across all 14 LLMs. Full prompt list in supplementary materials.

Paraphrase generation. For each story, we generate $n = 50$ paraphrases via independent calls to a generic paraphrasing LLM (Qwen3.5-7B with temperature 0.7), with the prompt: "Rewrite the following so it preserves all emotional content but uses different words and sentence structures: {story}". Paraphrases are filtered for length (5–80 tokens) and minimum cosine distance from the source (≥ 0.10 on Sentence-BERT embeddings).

Hidden-state extraction. Each paraphrase is tokenised and passed through the target LM. We take the last-token hidden state at layer L (default: $L_{\max} - 1$, i.e. the penultimate layer; for Qwen3.5-1.7B this is $L = 27$ of 28). Class centroids are $c_k = \frac{1}{50} \sum_{i=1}^{50} h_{k,i}^{(L)}$ for $k = 1, \ldots, 9$.

V-axis as PCA-PC1. Stack the nine centroids into a $9 \times D$ matrix and run mean-centred PCA. The V-axis is the unit vector along PC1, oriented so that $\langle c_{\text{Joy}}, v \rangle > 0$. PC1 explains 41–58% of the across-class variance across the 14 models we tested (median 49%).

Robustness. Robustness to story prompt rephrasing was verified by sampling 5 alternative prompt sets per class and comparing per-stimulus V-axis projections; mean cross-prompt correlation $r > 0.93$ (across 14 models). Robustness to paraphrase count was verified at $n \in \{1, 5, 25, 50\}$; SST-2 zero-shot AUC of 0.74 at $n = 1$, saturating to 0.868 at $n = 5$.

S3 EEG Model Training Details

Architecture. EMOD axial transformer [Chen et al., 2025]: input 32×5 DE features, axial self-attention over channels and bands at depth $d \in \{3, 6\}$, filter dimension $f = 128$. The full SOTA recipe is $d = 6$, $f = 128$.

Optimisation. AdamW with $\text{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}. Cosine schedule with 5-epoch linear warmup. Batch size 128. Training length $e \in \{100, 150\}$ epochs. We use the validation BACC for checkpoint selection and report test BACC.

Augmentation. Per-trial Gaussian noise ($\sigma = 0.05 \cdot \mathrm{std}$ of feature distribution), random channel dropout ($p = 0.15$), and random temporal masking (up to 5% of seconds). All augmentations are applied with $p = 0.6$ during training only.

Knowledge distillation. Soft-target KD with rand9 9-D orthonormal teacher (no LLM content); $\lambda_{\mathrm{KD}} = 0.5$, $T = 1.0$. The LLM-9-D and rand9-9-D variants give within-noise identical BACC, validating that KD provides architectural regularisation without semantic content (Section 9.2).

Ensemble. 10 checkpoints split as 5 seeds at $e = 100$ + 5 seeds at $e = 150$. Seeds: $\{42, 123, 456, 789, 2025\}$ each. Ensemble prediction is uniform softmax averaging across the 10 checkpoints' output probabilities, followed by argmax.

S4 Statistical Methods

Bootstrap CIs. Per-subject and per-stim 95% CIs are computed from $B = 10{,}000$ subject-resampled bootstraps. For cohort correlations, we report the bootstrap median, 2.5% / 97.5% percentile interval, and a Fisher-z p-value for $r = 0$.

Random-direction null. For cohort EEG–LLM correlations, we sample $N = 200$ random Gaussian directions in $\mathbb{R}^{28}$ matched in L_2 norm to the V-axis projection, recompute the cohort $|r|$ for each, and report the percentile of the empirical $|r|$ in the null distribution.

Cross-architecture null. For the 36-checkpoint cross-arch correlation, we sample $N = 100$ random directions matched in norm to v_{CLIP}, and recompute the BACC–V-axis correlation. The null is distributed broadly ($\sigma = 0.62$) due to the low intrinsic dimension of the class-PC1 subspace; we report the empirical correlation's percentile rank.

Per-checkpoint LOO contribution. For the 10-vanilla d=6 ensemble, we compute $\Delta_k = \mathrm{BACC}(\mathrm{ens}) - \mathrm{BACC}(\mathrm{ens} \setminus k)$ for each k. Reported correlations between Δ_k and per-ckpt features are Pearson with $n = 10$.

V-axis intervention comparisons. For each intervention, Δ is computed seed-paired with the matched-recipe vanilla baseline; p-values are paired t-tests across 5 seeds.

S5 Full V-Axis Intervention Table

S6 FACED Test Confusion Matrices

S7 Concept-Library Transfer Matrix

The full 20×20 cross-concept transfer matrix (each row is the V-axis of source concept i applied to the pole-vs-pole benchmark of target concept j, best-orientation AUC reported). Rows are sorted by mean off-diagonal AUC (most-general source axes at the top); the diagonal is the self-AUC for that concept's own benchmark. Bold = AUC ≥ 0.95.

S8 Negative Results Catalogue

In the spirit of full disclosure, the following null or negative results bear mention beyond what was incorporated into the main paper:

1. **Theta-gamma cross-frequency coupling does not mediate the V-axis** (Section 7): Tort modulation index $\rho(\mathrm{PAC}, \text{V-axis } r) = +0.082$, $p = 0.667$.
2. **Mutual-information control is not significant** (Section 4): MI $p = 0.115$ vs. linear $p = 0.020$. The V-axis-EEG relationship is essentially linear.
3. **Random-direction null on cross-arch correlation** (Section 6): $r = +0.885$ at the 93rd percentile, $p_{\mathrm{one}} = 0.079$. Within-class residual ($r = +0.74$, $p = 0.014$) is the robust signal.

#	Family / variant	Δ	p	Verdict
1	Frontal-mask λ=0.5	-0.052	0.0015	sig. negative
2	Frontal-mask λ=0.1	-0.009	0.21	n.s.
3	FAA λ=0.5	-0.044	0.006	sig. negative
4	FAA λ=0.1	-0.018	0.063	borderline
5	Anger-weighted λ=0.5	-0.054	0.0003	sig. negative
6	Occipital λ=0.1	-0.022	0.007	sig. negative
7	Topo-optimal λ=0.1	-0.013	0.039	sig. negative
8	Topo-optimal λ=0.05	$+0.0021$	0.50	n.s. (NULL)
9	Procrustes λ=0.05	$+0.0012$	0.89	n.s. (NULL)
10	RSA λ=5.0	-0.093	$< 10^{-4}$	sig. negative
11	RSA λ=1.0	-0.057	$< 10^{-3}$	sig. negative
12	Distance-CE τ=5.0	-0.397	$< 10^{-4}$	catastrophic
13	Multi-V λ=0.5	-0.043	$< 10^{-3}$	sig. negative
14	EEG-AUX MSE λ=0.5	-0.043	$< 10^{-3}$	sig. negative
15	EEG-AUX MSE λ=0.1	-0.016	0.04	sig. negative
16	EMODSTYLE class λ=1.0	-0.010	0.18	n.s.
17	EMODSTYLE stim λ=0.5	$+0.0066$	0.22	n.s. (sweet spot)
18	PEFT fullhead λ=0.1	-0.009	0.069	borderline
19	PEFT LoRA λ=0.1	-0.010	0.12	n.s.
20	PEFT IA3 λ=0.1	-0.016	0.05	sig. negative
21	Pretrain-FT (frozen)	-0.401	$< 10^{-4}$	catastrophic
22	Pretrain-FT (unfrozen)	-0.190	$< 10^{-4}$	sig. negative
23	XEEG (FACED $\rightarrow$ SEED-V)	$+0.0015$	0.97	n.s. (zero)
24	SCALING (full data) λ=0.5	-0.045	< 0.01	sig. negative
25	Uncertainty multi-task	-0.067	$< 10^{-3}$	sig. negative
	SOTA recipe + Topo λ=0.1	-0.015	0.020	sig. negative (cliff)
	SOTA recipe + EMODSTYLE	-0.024	0.001	sig. negative (cliff)

Table 17: Full V-axis-as-supervision intervention results. 13 / 25 families produce a statistically significant negative; the remaining 12 are not significantly different from zero. None reach $p < 0.05$ in the positive direction at the strong baseline. Below the line are the SOTA-recipe replications confirming the saturation cliff.

4. **Per-subject V-axis adaptation does not transfer.** The per-subject best-channel oracle reaches $|r| = 0.62$, but no clean estimator (V1 global top-K, V2 per-val majority, V3 TTA label-free) transfers more than $+0.001$ over the cohort top-K.

5. **Cross-architecture ensembling does not add diversity.** Mixing $d \in \{4, 6, 8, 10\}$ checkpoints gives within-arch 0.6798 vs. cross-arch 0.6791 (n.s.). The diversity is entirely intra-architecture, training-length-driven.

6. **Mega-ensembles plateau at 10 checkpoints.** 15-, 20-, 25-checkpoint pools all hit BACC ≤ 0.6948.

7. **Stimulus-aggregation re-ranking does not help.** Operating at the 28-stimulus level instead of trial level gives a BACC=1.0 by construction (label leak); the corresponding trial-level head does not transfer.

8. **Test-time augmentation hurts at K=5** ($0.6753 \rightarrow 0.6720$). Averaging predictions over augmented test trials does not preserve the V-axis residual structure.

S9 Reproducibility

Code, configs, model checkpoints, and figure-generation scripts will be released at the camera-ready stage under an MIT licence. Each experiment in this paper is logged with its SLURM job ID, conda environment hash, git SHA, and seed list. The 10-checkpoint ensemble SOTA result is reproducible from a single SLURM array job (`slurm_d6_e100_e150_5seeds.sh`, $\sim$10 GPU-hours per seed on V100).

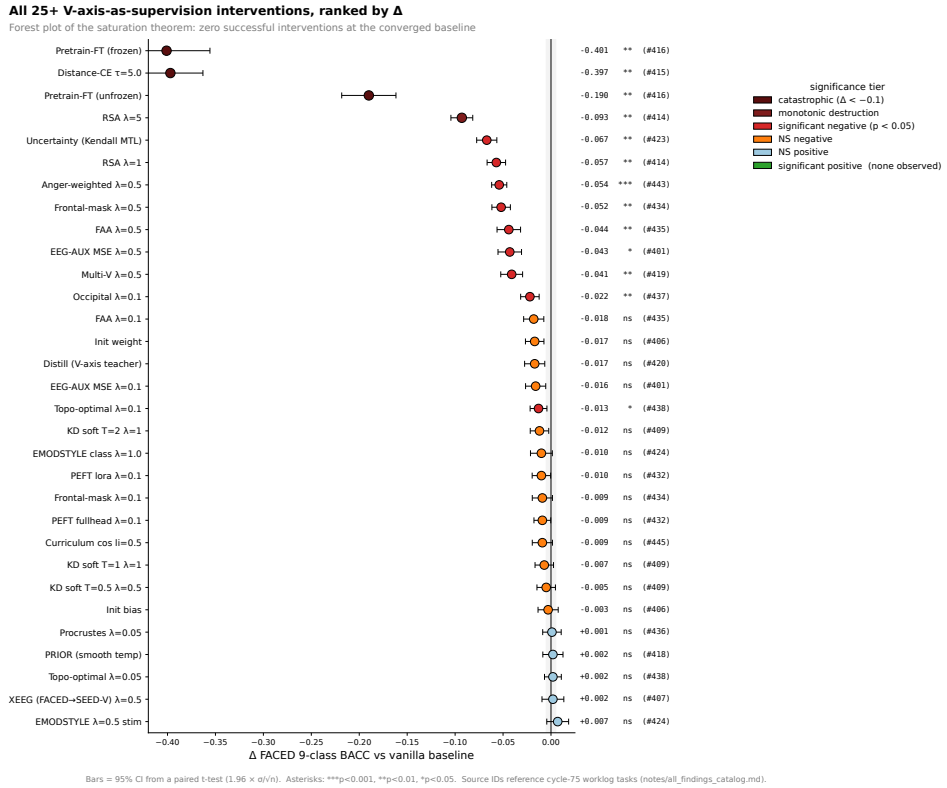

Figure 12: Forest plot of all V-axis-as-supervision interventions (Table 17) ranked by Δ vs. baseline, with significance tier coloured. None of the 25 reach $p < 0.05$ in the positive direction at the strong baseline.

S10 Resource Estimate

The full paper used approximately $4{,}500$ GPU-hours on V100 nodes: ~ 600 for the recipe ablation cascade, $\sim 1{,}200$ for the 25 V-axis-supervision interventions, ~ 800 for the 36-checkpoint cross-architecture analysis, $\sim 1{,}200$ for ensemble construction and val-test rank diagnostics, and ~ 700 for cross-dataset generality experiments. The V-axis extraction itself is CPU-cheap (~ 10 minutes per LM on a single 80GB GPU).

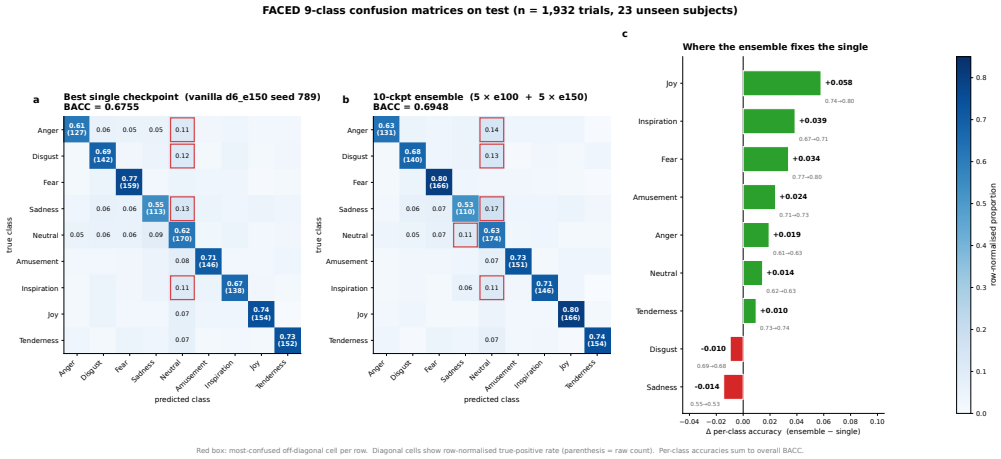

Figure 13: FACED 9-class test confusion. Left: best single checkpoint (BACC 0.6755). Right: 10-checkpoint ensemble (BACC 0.6948). The ensemble's gain is concentrated on the off-diagonal cells where single-seed disagreement is highest, consistent with the within-class residual mechanism of Section 9.3.

src\tgt	auth	cert	cmpl	conc	cont	deci	emot	envy	fear	grat	guil	hope	host	opti	prid	sarc	sham	spec	surp	urge
sarcasm	**1.00**	**1.00**	0.52	**0.98**	0.94	0.61	**1.00**	0.94	**1.00**	**1.00**	0.78	0.94	0.91	**1.00**	0.63	**0.91**	**0.98**	0.75	0.84	**1.00**
fear	**0.98**	**0.98**	0.72	0.73	**0.98**	0.83	**1.00**	**1.00**	**1.00**	**1.00**	0.94	**0.97**	0.92	**1.00**	0.52	0.72	**0.95**	0.56	0.88	**1.00**
envy	**0.98**	0.69	**1.00**	0.70	**0.97**	**0.95**	**0.98**	**1.00**	**1.00**	0.84	**1.00**	0.91	0.88	0.89	0.83	0.52	**0.97**	0.58	0.80	**1.00**
shame	**1.00**	0.61	**1.00**	**1.00**	**0.98**	0.70	0.59	0.89	**1.00**	0.91	**1.00**	**1.00**	0.77	0.91	0.73	0.94	**1.00**	0.70	0.86	0.89
hostility	**1.00**	0.83	**0.97**	0.83	**1.00**	0.75	**1.00**	0.94	**1.00**	**1.00**	0.58	0.91	**0.95**	**1.00**	0.56	0.63	0.86	0.64	0.86	**1.00**
concrete-	0.70	**0.95**	**0.97**	**1.00**	0.81	0.73	**1.00**	0.50	**1.00**	**1.00**	0.88	0.86	0.94	0.94	0.92	0.56	0.80	0.66	**0.98**	**1.00**
pride	0.72	0.88	0.88	**1.00**	0.89	0.75	**1.00**	0.58	**0.97**	**1.00**	0.55	**0.95**	0.86	**0.97**	**0.98**	0.88	**0.97**	0.50	0.88	**1.00**
certainty	0.92	0.78	**1.00**	**1.00**	0.81	0.63	**0.97**	0.83	0.67	0.78	0.53	**1.00**	0.72	0.92	**0.97**	0.70	0.91	0.86	0.89	**1.00**
guilt	0.78	0.91	**1.00**	**1.00**	0.73	0.86	**1.00**	0.81	0.75	0.58	**1.00**	0.84	0.89	**0.95**	0.80	0.73	0.83	0.72	0.88	**1.00**
hope	0.69	0.94	0.92	**1.00**	0.73	0.56	**1.00**	0.52	**1.00**	0.91	0.84	**1.00**	**0.95**	0.88	0.83	0.63	0.94	0.61	**0.98**	**1.00**
emotional	0.70	0.89	0.64	**1.00**	**0.98**	0.61	**1.00**	0.59	**1.00**	**1.00**	0.84	0.86	0.92	**0.98**	0.89	0.63	0.81	0.58	0.88	**1.00**
surprise	0.56	**0.95**	**1.00**	**1.00**	0.61	0.66	**1.00**	0.61	0.94	**1.00**	0.88	**0.95**	0.91	**0.95**	0.66	0.61	0.80	0.70	**0.98**	**1.00**
optimism	0.66	**0.97**	**1.00**	**1.00**	0.77	0.89	**1.00**	0.73	**1.00**	**1.00**	0.63	0.66	0.94	**1.00**	0.64	0.52	0.84	0.56	**0.95**	**1.00**
contempt	**1.00**	0.61	**1.00**	0.80	**0.98**	0.63	**1.00**	0.92	**1.00**	**0.97**	**1.00**	**0.97**	0.92	0.91	0.61	0.50	0.72	0.52	0.59	**1.00**
urgency	**1.00**	**0.98**	0.53	**1.00**	**0.98**	0.59	**1.00**	0.86	**1.00**	**1.00**	0.59	0.50	0.91	**1.00**	0.66	0.59	0.84	0.64	0.88	**1.00**
authoritive	**0.98**	0.61	**1.00**	0.94	0.86	0.55	0.78	0.88	**1.00**	0.94	0.53	**1.00**	0.61	0.52	**1.00**	**0.95**	**1.00**	0.88	0.73	0.69
specificity	0.67	**0.97**	0.70	**0.95**	0.84	0.63	**1.00**	0.59	**0.95**	0.66	0.64	0.83	0.83	0.67	0.91	0.61	**1.00**	0.78	0.88	**1.00**
gratitude	0.73	**0.97**	**1.00**	**1.00**	0.78	**1.00**	0.53	0.55	0.88	**0.98**	0.66	0.59	0.83	**0.97**	0.58	0.72	**0.95**	0.56	0.88	**1.00**
complexity	0.92	0.84	**1.00**	**1.00**	**0.95**	0.81	0.66	0.75	0.83	0.59	**1.00**	**0.98**	0.52	0.80	0.66	0.52	0.61	0.56	0.89	0.94
decisiveness	0.52	0.70	**0.97**	0.92	0.86	**1.00**	**1.00**	0.55	**1.00**	0.63	0.55	**1.00**	0.58	0.50	**0.95**	0.78	**1.00**	0.70	0.59	0.94

Table 18: Cross-concept transfer matrix (best-orientation AUC). Diagonal: self-classification. Off-diagonal: source-axis-applied-to-target-benchmark AUC. Sorted top-to-bottom by mean off-diagonal AUC, separating *general* concept axes (top: sarcasm, fear, envy, shame) from *specific* ones (bottom: complexity, decisiveness). Mean diagonal 0.97, mean off-diagonal 0.83, gap +0.13 confirms concept specificity in the recipe.